Norm-Transferring Skip Connections and Radial Foliation of Conserved Quantities in Dynamical Autoencoders

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Abstract

Standard normalization layers such as LayerNorm and RMSNorm project activations onto a sphere of fixed radius, discarding the input norm $\|x\|_p$ at every block. We study **SkipNorm**, a norm-transferring skip connection $\hat{y} = \|x\|_p \, f(x) / \|f(x)\|_p$ that decouples angular feature transformation from radial magnitude transport, alongside its tangent-space retraction **Geodesic SkipNorm**, which eliminates the antipodal sign-flip instability of unprojected normalized residuals when a residual update opposes the input direction. First, we establish the exact differential geometry of SkipNorm and stabilized normalizers: we prove that for $p = 2$, the Jacobian transpose of SkipNorm pulls the output radial unit covector back to the exact input radial unit covector ($J^\top u_{\text{out}} = u_{\text{in}}$) across arbitrary dimensions, while explicit counterexamples show that $Ju_{\text{in}} = u_{\text{out}}$ and unit radial singular values fail whenever radial-to-angular derivatives $B = (\partial u_{\text{out}}/\partial r)\,\|x\|_2$ are nonzero. We derive the exact singular value spectra of ε-stabilized RMSNorm and LayerNorm, showing that their radial singular value is suppressed by $\mathcal{O}(\varepsilon/r^3)$ rather than vanishing identically when $\varepsilon > 0$. Second, on the focusing 1D discrete nonlinear Schrödinger (DNLS) equation integrated by a converged reference solver without artificial post-step normalization, Geodesic SkipNorm preserves the discrete L_2 soliton norm to machine precision ($\sim 4.1 \times 10^{-15}$) over 250 rollout steps at 7.4× lower latency than state-dependent SO(32) Cayley transitions. Third, we prove a conditional **Radial Foliation Theorem**: when an injective encoder intertwines a continuous flow on connected level sets with a norm-preserving latent transition and orbits are topologically dense in each connected level set, the latent radius is strictly constant on each level set ($\mathrm{CV}_{\mathcal{O}}(h) = 0$) and provides a continuous, strictly monotone coordinate on regular level intervals. Crucially, we prove and empirically verify two topological obstructions: a **Jordan–Brouwer Multi-Well Obstruction** prohibiting simultaneous origin-enclosing embeddings of disconnected sub-barrier basins under global homeomorphisms, and a **Toroidal Multi-Invariant Obstruction** on $\mathbb{T}^K$ ($K \geq 2$) resolved by generalizing SkipNorm to K-block direct sums $\bigoplus_{k=1}^{K} \mathbb{R}^{d_k}$.

1 Introduction

Deep residual networks, recurrent architectures, and neural differential equations rely on normalization and skip connections to stabilize signal and gradient propagation across depth and time (He et al., 2016; Ba et al., 2016; Zhang & Sennrich, 2019). Yet standard normalizers—including LayerNorm (Ba et al., 2016), RMSNorm (Zhang & Sennrich, 2019), Cosine Normalization (Luo et al., 2018), and hyperspherical architectures such as nGPT (Loshchilov et al., 2024)—achieve angular conditioning by resetting the Euclidean norm of the activation vector to a fixed constant or learned parameter. In doing so, they discard the input magnitude $\|x\|_2$ at every layer. While discarding radial scale is benign when scale is nuisance variation, in physical forecasting, conservative dynamical systems, and representation learning of extensive state variables, the norm carries essential energy, mass, or amplitude information.

A natural resolution is to normalize the nonlinear branch $f(x)$ onto the unit sphere and transfer the input norm $\|x\|_p$ multiplicatively via a scalar skip connection:

$$\mathrm{SkipNorm}_p(x; f) \coloneqq \frac{f(x)}{\|f(x)\|_p} \|x\|_p, \qquad x \in \mathbb{R}^{d_{\text{in}}} \setminus \{0\}, \quad f(x) \in \mathbb{R}^{d_{\text{out}}} \setminus \{0\}. \tag{1}$$

Despite its simplicity, the exact differential geometry of Eq. (1) and its consequences for dynamical representation learning are surprisingly subtle. First, it is tempting to assume that because $\|\hat{y}\|_2 = \|x\|_2$, the Jacobian $J = \partial\hat{y}/\partial x$ maps the input radial unit vector $u_{\text{in}} = x/\|x\|_2$ to the output radial unit vector $u_{\text{out}} = \hat{y}/\|\hat{y}\|_2$ and possesses a unit singular value $\sigma = 1$. As we prove in Section 3, both forward statements are *false* in general unless the radial-to-angular derivative $B \coloneqq (\partial u_{\text{out}}/\partial r)\|x\|_2$ vanishes; what holds unconditionally for $p = 2$ across arbitrary dimensions $d_{\text{in}}, d_{\text{out}}$ is the **cotangent radial pullback identity** $J^\top u_{\text{out}} = u_{\text{in}}$, which guarantees $\sigma_{\max}(J) \geq 1$ and exact unit-norm backward propagation of radial gradients across arbitrary depth. Second, when Eq. (1) is combined with a residual branch $y = x + \alpha f(x)$ (as in normalized residual steps), large opposing radial components $\alpha\langle f(x), u_{\text{in}}\rangle < -\|x\|_2$ trigger a discontinuous **antipodal sign-flip instability** ($\hat{y} \approx -x$). Projecting $f(x)$ onto the tangent space $T_x\mathbb{S}^{d-1}(\|x\|_2)$ prior to spherical retraction yields **Geodesic SkipNorm**, which eliminates the antipodal flip unconditionally and guarantees a strictly first-order tangential update for all step sizes $\alpha > 0$.

Beyond single-layer conditioning, norm-preserving transitions play a foundational role in latent-space dynamical models (E, g, D), where an encoder $E : \mathcal{M} \to \mathbb{R}^{d_z}$ maps observed trajectories x_t into a latent space evolved by a norm-preserving map $g : \mathbb{R}^{d_z} \to \mathbb{R}^{d_z}$ ($\|g(z)\|_2 = \|z\|_2$). When an autoencoder is trained to predict multi-step rollouts without access to ground-truth conserved quantities, how does the interplay between the norm-preserving latent transition g and the topology of the observed phase space $\mathcal{M}$ organize the latent space? Rather than claiming an unconditional "invariant discovery machine," we formulate and prove exact *conditional* theorems alongside explicit *topological and dynamical counterexamples* that delineate precisely when latent radial foliation succeeds and when it must fail.

Summary of Contributions.

1. **Exact Jacobian Geometry, Stabilized Normalizer Spectra, and Tangent Retraction (Section 3):** We prove the exact block decomposition of the SkipNorm Jacobian across arbitrary dimensions $d_{\text{in}}, d_{\text{out}}$ (Proposition 1), establishing the unconditional cotangent pullback $J^\top u_{\text{out}} = u_{\text{in}}$ while providing explicit analytical counterexamples (2×2 logarithmic-phase twist with singular values $\{\varphi, \varphi^{-1}\}$ and $1 \to 2$ spiral with $\sigma = \sqrt{2}$) whenever $B \neq 0$. We derive the exact singular value spectra of ε-stabilized RMSNorm and LayerNorm (Proposition 2) and prove that tangent-projected Geodesic SkipNorm eliminates the antipodal sign-flip instability of unprojected normalized residuals (Proposition 3).

2. **Projection-Free Unitary Rollouts on 1D DNLS (Section 4):** On the focusing 1D discrete nonlinear Schrödinger (DNLS) equation integrated by a 10×-substepped RK4 reference solver *without* post-step normalization (intrinsic solver drift 1.32×10^{-11}), Geodesic SkipNorm preserves the discrete L_2 soliton norm to machine precision (4.09×10^{-15}) over 250 steps across 5 seeds while running 7.4× faster than state-dependent SO(32) Cayley orthogonal steps.

3. **Conditional Radial Foliation Theorem and Orbit-Density Control (Section 5):** We prove that if a continuous encoder E intertwines a flow with a norm-preserving latent map g and orbits are topologically dense in each connected level set $\mathcal{O}_c$, then $\|E(x)\|_2 = \psi(c)$ is constant on each $\mathcal{O}_c$, and under global injectivity ψ is strictly monotone on regular level intervals (Theorem 1). A resonant $1:1$ 2D harmonic oscillator control in $\mathbb{R}^4$ demonstrates why topological orbit density is indispensable: exact intertwining with $g \in \mathrm{SO}(4)$ achieves zero within-orbit radial variation ($\mathrm{CV}_{\mathcal{O}}(h) = 4.17 \times 10^{-16}$) yet exhibits a 50% radius spread across orbits on the same energy shell $H = 1$.

4. **Jordan–Brouwer Multi-Well and Toroidal $\mathbb{T}^K$ Obstructions (Sections 6–7):** Using the Jordan–Brouwer separation theorem, we prove that no continuous injective encoder $E : U \to \mathbb{R}^N$

can map two disjoint sub-barrier level components $\mathcal{O}_c^{(1)}, \mathcal{O}_c^{(2)} \cong \mathbb{S}^{N-1}$ onto concentric spheres around the origin while keeping both basins origin-enclosing (Theorem 2), and analyze the non-concentric off-origin bypass via an explicit translated-encoder control $E_a(x) = ae_1 + x$ with $\mathcal{O}(a^{-1})$ radial variation. Finally, we prove why a single spherical norm constraint cannot identify $K \geq 2$ independent invariants on Liouville–Arnold tori $\mathbb{T}^K$ and resolve this obstruction via **K-Block Toroidal Geodesic SkipNorm** on $\bigoplus_{k=1}^{K} \mathbb{R}^{d_k}$, raising held-out (I_1, I_2) readout R^2 from 0.146 (1-scalar ceiling) to 0.9993 (Theorem 3).

2 Related Work

Normalization Layers and Magnitude–Direction Disentanglement. Standard normalization layers project representations onto a fixed-radius manifold: LayerNorm (Ba et al., 2016) centers and scales activations onto the $(d-2)$-sphere in the zero-mean hyperplane, while RMSNorm (Zhang & Sennrich, 2019) and Cosine Normalization (Luo et al., 2018) project onto $\mathbb{S}^{d-1}(\sqrt{d})$ or $\mathbb{S}^{d-1}(1)$. Weight Normalization (Salimans & Kingma, 2016) decouples weight vector norms from directions in parameter space, and hyperspherical VAEs (Davidson et al., 2018; Xu & Durrett, 2018) model directional latent distributions via von Mises–Fisher densities. In modern Transformers, Pre-LN residual streams suffer from unchecked activation norm growth with depth, prompting nGPT (Loshchilov et al., 2024) to constrain hidden states and updates to the unit hypersphere via spherical linear interpolation, and Dynamic Tanh (DyT) (Zhu et al., 2025) to replace normalization with bounded elementwise squashing. Unlike unit-sphere normalizers that discard $\|x\|_2$, SkipNorm transfers the input norm $\|x\|_p$ across arbitrary nonlinear branches $f : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^{d_{\text{out}}}$.

Dynamical Isometry and Orthogonal/Unitary Transitions. Preserving singular values near unity across deep or recurrent compositions (*dynamical isometry*; Saxe et al., 2014; Pennington et al., 2017) motivated unitary and orthogonal RNNs parameterized via products of structured matrices (Arjovsky et al., 2016), exponential maps (Lezcano-Casado & Martínez-Rubio, 2019), or Cayley transforms $Q(S) = (I - S)(I + S)^{-1}$ (Helfrich et al., 2018). However, linear orthogonal maps $Q \in O(d)$ require equal input and output dimensions ($d_{\text{in}} = d_{\text{out}}$), cannot incorporate state-dependent nonlinearities without recomputing an $\mathcal{O}(d^3)$ matrix inverse or factorization at every step, and act globally on $\mathbb{R}^d$. Geodesic SkipNorm preserves Euclidean norms exactly in $\mathcal{O}(d)$ vector arithmetic while permitting arbitrary nonlinear tangent vector fields.

Physics-Informed Priors and Conserved-Quantity Discovery. Hamiltonian Neural Networks (HNNs; Greydanus et al., 2019), Symplectic ODE-Nets (Zhong et al., 2020), and Lagrangian Neural Networks (Cranmer et al., 2020) enforce conservation laws by hard-coding canonical symplectic or Euler–Lagrange structures on known phase-space coordinates (q, p). Conversely, invariant-discovery methods seek conserved quantities from trajectory data without canonical coordinate assumptions: Siamese and manifold-learning approaches (Wetzel et al., 2020; Liu & Tegmark, 2021) contrast same-orbit and different-orbit pairs, PoincareFlows (Barajas-Solano & Tartakovsky, 2025) learn volume-preserving flows, and COMET (Otto et al., 2023) penalizes the orthogonal complement of candidate invariant gradients along observed velocities. Our work bridges norm-preserving architectural primitives with invariant manifold theory: we show how intertwining an autoencoder with a spherical or toroidal SkipNorm transition organizes conserved quantities into latent radii under explicit topological conditions, and we prove exact topological obstructions (via invariance of domain and Jordan–Brouwer separation; Brouwer, 1911; Hatcher, 2002) when level sets are disconnected or toroidal (Arnold, 1989).

3 Differential Geometry of SkipNorm and Stabilized Normalizers

We first analyze the exact first-order differential geometry of SkipNorm, compare its Jacobian spectrum against ε-stabilized RMSNorm and LayerNorm, and introduce tangent-projected Geodesic SkipNorm. All proofs and explicit counterexample derivations are provided in Appendix A.

3.1 Exact Jacobian Decomposition and Cotangent Radial Pullback

Let $x \in \mathbb{R}^{d_{\text{in}}} \setminus \{0\}$ with radius $r = \|x\|_2 > 0$ and unit radial direction $u_{\text{in}} = x/r \in \mathbb{S}^{d_{\text{in}}-1}$. Let $f : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^{d_{\text{out}}}$ be C^1 on a neighborhood of x with $y = f(x) \neq 0$, output unit direction $u_{\text{out}} = y/\|y\|_2 \in \mathbb{S}^{d_{\text{out}}-1}$, and unstabilized Euclidean SkipNorm map

$$F(x) \coloneqq \frac{f(x)}{\|f(x)\|_2} \|x\|_2 = r\, u_{\text{out}}(x), \qquad \|F(x)\|_2 = \|x\|_2 . \tag{2}$$

In stabilized numerical implementations with $\varepsilon > 0$, we write $F_\varepsilon(x) = f(x)\sqrt{\|x\|_2^2 + \varepsilon}/\sqrt{\|f(x)\|_2^2 + \varepsilon}$. Differentiating $\|u_{\text{out}}(x)\|_2^2 \equiv 1$ gives $u_{\text{out}}^\top D_x u_{\text{out}}(x) = 0$, so the image of $D_x u_{\text{out}}(x) = \|y\|_2^{-1} (I_{d_{\text{out}}} - u_{\text{out}} u_{\text{out}}^\top) D_x f(x)$ lies strictly in the tangent hyperplane $T_{u_{\text{out}}} \mathbb{S}^{d_{\text{out}}-1} = u_{\text{out}}^\perp$. Decomposing $\mathbb{R}^{d_{\text{in}}} = \text{span}(u_{\text{in}}) \oplus u_{\text{in}}^\perp$ and $\mathbb{R}^{d_{\text{out}}} = \text{span}(u_{\text{out}}) \oplus u_{\text{out}}^\perp$ reveals the exact structure of the Jacobian $J(x) = D_x F(x)$.

Proposition 1 (Exact Jacobian Geometry of Euclidean SkipNorm)**.** *Let $F : \mathbb{R}^{d_{\text{in}}} \setminus \{0\} \to \mathbb{R}^{d_{\text{out}}}$ be given by Eq. (2) at $x \neq 0$ with $f(x) \neq 0$. Define the radial-to-angular sensitivity vector $B(x) \in u_{\text{out}}^\perp$ and the tangential-to-angular operator $A_\perp(x) : u_{\text{in}}^\perp \to u_{\text{out}}^\perp$ by*

$$B(x) \coloneqq r(D_x u_{\text{out}}(x)) u_{\text{in}} = r \frac{\partial u_{\text{out}}(r u_{\text{in}})}{\partial r}\bigg|_{r=\|x\|_2} \in u_{\text{out}}^\perp, \qquad A_\perp(x) \coloneqq r D_x u_{\text{out}}(x)|_{u_{\text{in}}^\perp} . \tag{3}$$

Then:

- ***(a) Full Jacobian and Block Matrix Representation:*** *In any orthonormal bases $\{u_{\text{in}}, v_1, \ldots, v_{d_{\text{in}}-1}\}$ of $\mathbb{R}^{d_{\text{in}}}$ and $\{u_{\text{out}}, w_1, \ldots, w_{d_{\text{out}}-1}\}$ of $\mathbb{R}^{d_{\text{out}}}$,*

$$J(x) = u_{\text{out}} u_{\text{in}}^\top + B(x) u_{\text{in}}^\top + A_\perp(x) P_{u_{\text{in}}^\perp}, \qquad [J(x)] = \begin{pmatrix} 1 & 0_{1\times(d_{\text{in}}-1)} \\ b & \tilde{A} \end{pmatrix} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}, \tag{4}$$

  where $b \in \mathbb{R}^{d_{\text{out}}-1}$ and $\tilde{A} \in \mathbb{R}^{(d_{\text{out}}-1)\times(d_{\text{in}}-1)}$ represent $B(x)$ and $A_\perp(x)$.

- ***(b) Unconditional Cotangent Radial Pullback:*** *Across all dimensions $d_{\text{in}}, d_{\text{out}} \geq 1$ and all C^1 branches f, the transpose Jacobian satisfies*

$$J(x)^\top u_{\text{out}} = u_{\text{in}}, \qquad \left\|J(x)^\top u_{\text{out}}\right\|_2 = 1, \qquad \sigma_{\max}(J(x)) \geq 1. \tag{5}$$

  Furthermore, for any composition $F_{\text{total}} = F_L \circ \cdots \circ F_1$ of L SkipNorm layers with $x_\ell = F_\ell(x_{\ell-1}) \neq 0$, the end-to-end Jacobian $J_{\text{total}} = \prod_{\ell=1}^{L} J_\ell$ satisfies $J_{\text{total}}^\top u_L = u_0$ and $\left\|J_{\text{total}}^\top u_L\right\|_2 = 1$ identically.

- ***(c) Conditional Forward Pushforward and Unit Singular Value:*** *The forward radial pushforward equals $J(x)u_{\text{in}} = u_{\text{out}} + B(x)$ with $\|J(x)u_{\text{in}}\|_2 = \sqrt{1 + \|B(x)\|_2^2}$. Consequently, $J(x)u_{\text{in}} = u_{\text{out}}$ if and only if $B(x) = 0$ (which holds automatically whenever f is positively homogeneous of any degree $\gamma \in \mathbb{R}$, i.e., $f(\lambda x) = \lambda^\gamma f(x)$ for $\lambda > 0$). When $B(x) = 0$, the block matrix $[J(x)] = \text{diag}(1, \tilde{A})$ decouples orthogonally, so $\sigma = 1$ is an exact singular value of $J(x)$ with left/right singular vectors $(u_{\text{out}}, u_{\text{in}})$ and the remaining singular values are $\sigma(\tilde{A})$.*

- ***(d) Explicit Counterexamples when $B(x) \neq 0$:***

  (i) Equidimensional Twist ($d_{\text{in}} = d_{\text{out}} = 2$): *For $f(x) = R_{\log\|x\|_2} x$, where $R_\theta \in \text{SO}(2)$ is planar rotation by $\theta = \log\|x\|_2$, we have $\|f(x)\|_2 = \|x\|_2$ so $F(x) = f(x)$. At $x = e_1 = (1,0)^\top$, $B(e_1) = e_2 \neq 0$ and*

$$J(e_1) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \sigma(J(e_1)) = \left\{\varphi, \varphi^{-1}\right\} = \left\{\frac{\sqrt{5}+1}{2}, \frac{\sqrt{5}-1}{2}\right\} \approx \{1.6180, 0.6180\}.$$

  Thus $1 \notin \sigma(J(e_1))$ even though $d_{\text{in}} = d_{\text{out}} = 2$ and $\|F(x)\|_2 \equiv \|x\|_2$.

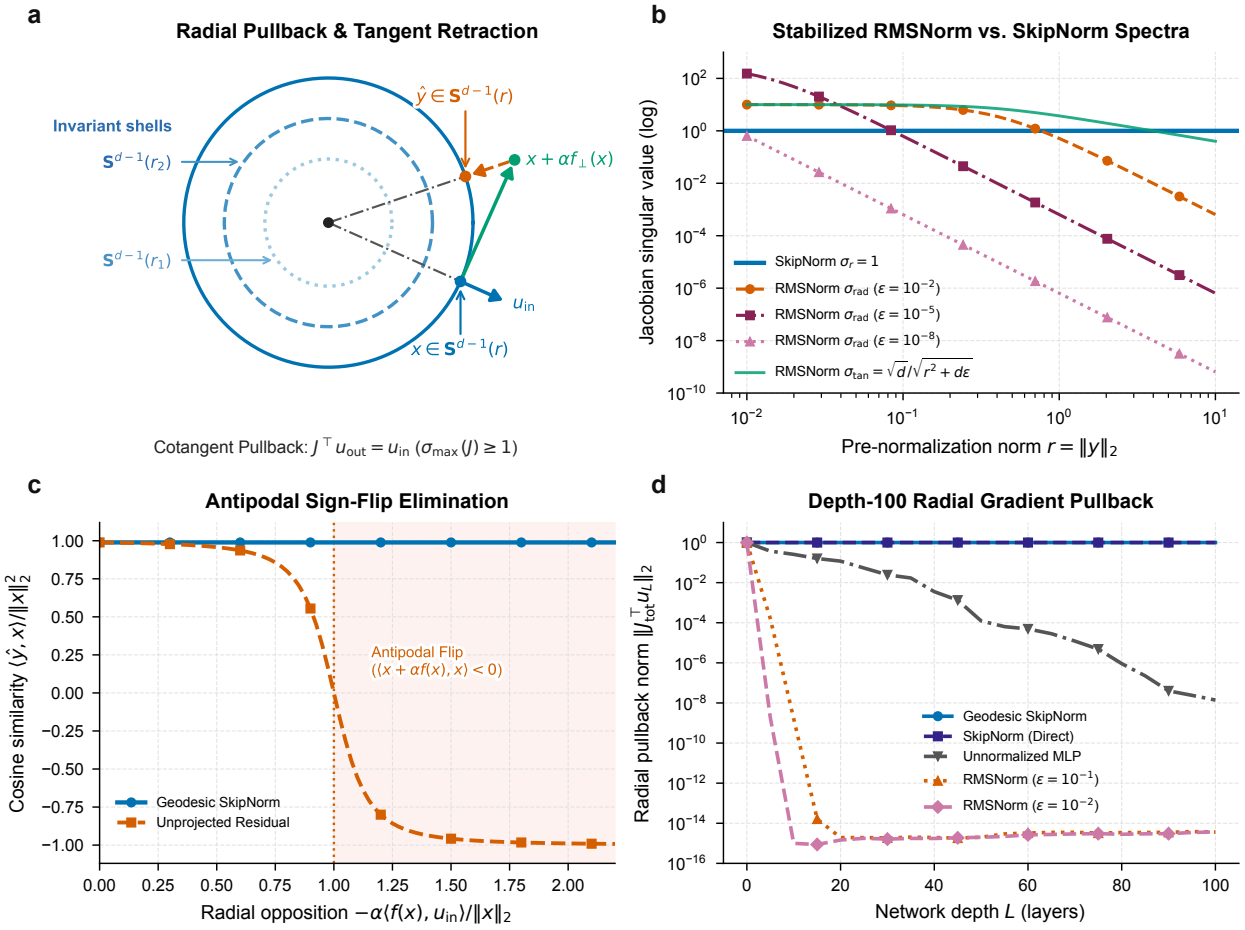


Figure 1: **Exact Jacobian geometry of SkipNorm, stabilized normalizer spectra, and tangent-space Geodesic retraction. (a)** Geometric decomposition on $\mathbb{S}^{d-1}(r)$: Geodesic SkipNorm projects $f(x)$ onto $T_x\mathbb{S}^{d-1}(r)$ before retracting onto the input sphere, guaranteeing the unconditional cotangent radial pullback $J^\top u_{\text{out}} = u_{\text{in}}$. **(b)** Exact singular values of ε-stabilized RMSNorm ($d = 16$) versus pre-normalization norm $r = \|y\|_2$: solid/dashed curves show Proposition 2 analytical formulas and markers show JAX float64 autodiff SVDs (max discrepancy 1.33×10^{-13}). **(c)** Cosine similarity $\langle \hat{y}, x \rangle / \|x\|_2^2$ versus radial opposition $-\alpha \langle f(x), u_{\text{in}} \rangle / \|x\|_2$: unprojected Projected Residual SkipNorm suffers an antipodal sign flip ($\hat{y} \to -x$) past threshold 1, whereas tangent-projected Geodesic SkipNorm remains smooth and flip-free. **(d)** Depth-100 backward radial gradient pullback $\left\|J_{\text{total}}^\top u_L\right\|_2$: SkipNorm and Geodesic SkipNorm preserve exact unit radial pullback (1.0000) across 100 layers, whereas stabilized RMSNorm attenuates radial gradients exponentially as $(\sigma_{\text{rad}}(\varepsilon))^L$ down to the float64 precision floor ($\sim 10^{-15}$).

(ii) Dimension-Changing Spiral ($d_{\text{in}} = 1, d_{\text{out}} = 2$): *For* $x = t > 0$ *and* $f(t) = (\cos \log t, \sin \log t)^\top$, $F(t) = t(\cos \log t, \sin \log t)^\top$ *satisfies* $\|F(t)\|_2 = t$. *At* $t = 1$, $J(1) = (1, 1)^\top$, *whose sole singular value is* $\sigma_1(J(1)) = \sqrt{2} \approx 1.4142 \neq 1$, *while* $J(1)^\top u_{\text{out}} = 1 = u_{\text{in}}$.

Proposition 1 clarifies a critical geometric distinction: norm preservation $\|F(x)\|_2 = \|x\|_2$ is a statement on the *cotangent bundle* ($d(\|F(x)\|_2) = d(\|x\|_2) \implies J(x)^\top u_{\text{out}} = u_{\text{in}}$), not on the forward pushforward $J(x)u_{\text{in}}$. Whenever a nonlinear layer $f(x)$ alters its output direction as the input radius $r = \|x\|_2$ varies ($B(x) \neq 0$), a radial perturbation $\delta x = \delta r\, u_{\text{in}}$ induces both a radial step $\delta r\, u_{\text{out}}$ and a tangential shear $\delta r\, B(x)$. Nevertheless, because the top row of $[J(x)]$ in Eq. (4) is $(1, 0)$ regardless of b, backward propagation of radial gradients ($J(x)^\top u_{\text{out}} = u_{\text{in}}$) and the rank decoupling $\text{rank}(J(x)) = 1 + \text{rank}(\tilde{A})$ hold unconditionally.

3.2 Exact Singular Value Spectra of ε-Stabilized RMSNorm and LayerNorm

A common folklore claim is that RMSNorm and LayerNorm have a strictly zero radial singular value ($\sigma_{\text{rad}} = 0$) and rank deficiency of 1 or 2. While true for unstabilized maps ($\varepsilon = 0$), practical implementations always include a numerical stabilizer $\varepsilon > 0$ inside the square root. Proposition 2 establishes the exact singular value spectra for both $\varepsilon = 0$ and $\varepsilon > 0$.

Proposition 2 (Exact Rank and Stabilized Singular Value Spectra of SkipNorm, RMSNorm, and LayerNorm)**.** *Let $y \in \mathbb{R}^d \setminus \{0\}$ with $r = \|y\|_2 > 0$ and $u = y/r$. For LayerNorm, let $\tilde{y} = (I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top)y \neq 0$ with $r_c = \|\tilde{y}\|_2 > 0$ and $u_c = \tilde{y}/r_c \in \mathbf{1}^\perp$.*

(a) Exact Rank Relation for Unstabilized Normalizers *($\varepsilon = 0$): For $y = f(x)$ with $A = D_x f(x) \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and $P_\perp = I_{d_{\text{out}}} - u_{\text{out}} u_{\text{out}}^\top$, the tangential restriction is $A|_T \coloneqq P_\perp A P_{u_{\text{in}}^\perp}$. Then:*

$$\operatorname{rank}(J_{\text{SkipNorm}}(x)) = 1 + \operatorname{rank}(A|_T), \qquad \operatorname{rank}(J_{\text{RMSNorm},0}(x)) = \operatorname{rank}(P_\perp A). \tag{6}$$

In particular, $\operatorname{rank}(J_{\text{SkipNorm}}(x))$ depends on $\operatorname{rank}(A|_T)$ rather than $\operatorname{rank}(A)$: for $d_{\text{in}} = d_{\text{out}} = 2$ at $x = e_1$ with $u_{\text{out}} = e_1$ and $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, we have $\operatorname{rank}(A) = 1$ yet $A|_T = 0$, yielding $\operatorname{rank}(J_{\text{SkipNorm}}(e_1)) = 1 + 0 = 1 < 2$. Conversely, whenever $A \in \mathrm{GL}(d)$ is invertible ($d_{\text{in}} = d_{\text{out}} = d$), $\operatorname{rank}(A|_T) = d - 1$ so $\operatorname{rank}(J_{\text{SkipNorm}}(x)) = d$, whereas $\operatorname{rank}(J_{\text{RMSNorm},0}(x)) = d - 1$ and $\operatorname{rank}(J_{\text{LayerNorm},0}(x)) = d - 2$.

(b) Exact Spectrum of ε-Stabilized RMSNorm *($\varepsilon > 0$): The primitive $\operatorname{RMSNorm}_\varepsilon(y) = \sqrt{d}\, y / \sqrt{\|y\|_2^2 + d\varepsilon}$ has Jacobian*

$$D_y \operatorname{RMSNorm}_\varepsilon(y) = \frac{\sqrt{d}}{\sqrt{r^2 + d\varepsilon}}(I_d - uu^\top) + \frac{d^{3/2}\varepsilon}{(r^2 + d\varepsilon)^{3/2}} uu^\top. \tag{7}$$

Since $I_d - uu^\top$ and $uu^\top$ are mutually orthogonal symmetric projections, $D_y \operatorname{RMSNorm}_\varepsilon(y)$ is symmetric positive definite ($\operatorname{rank} = d$) for all $\varepsilon > 0$ with exact singular values:

$$\sigma_{\tan}(r;\varepsilon) = \frac{\sqrt{d}}{\sqrt{r^2 + d\varepsilon}} \quad (\textit{multiplicity } d-1), \qquad \sigma_{\text{rad}}(r;\varepsilon) = \frac{d^{3/2}\varepsilon}{(r^2 + d\varepsilon)^{3/2}} \quad (\textit{multiplicity } 1). \tag{8}$$

(c) Exact Spectrum of ε-Stabilized LayerNorm *($\varepsilon > 0$): The primitive $\operatorname{LayerNorm}_\varepsilon(y) = \sqrt{d}\, \tilde{y} / \sqrt{\|\tilde{y}\|_2^2 + d\varepsilon}$ has symmetric positive semidefinite Jacobian ($\operatorname{rank} = d - 1$) with exact singular values:*

$$\sigma_{\tan}(r_c;\varepsilon) = \frac{\sqrt{d}}{\sqrt{r_c^2 + d\varepsilon}} \ (\textit{mult. } d-2), \quad \sigma_{\text{rad}}(r_c;\varepsilon) = \frac{d^{3/2}\varepsilon}{(r_c^2 + d\varepsilon)^{3/2}} \ (\textit{mult. } 1), \quad \sigma_{\mathbf{1}} = 0 \ (\textit{mult. } 1). \tag{9}$$

Proposition 2(b) explains why ε-stabilized RMSNorm still behaves as a strong radial damper in standard training regimes: at typical pre-normalization norm $r = \sqrt{d}$ with $\varepsilon = 10^{-5}$, the radial singular value is $\sigma_{\text{rad}} \approx \varepsilon/(1+\varepsilon)^{3/2} \approx 10^{-5} \ll 1$, attenuating radial gradients by five orders of magnitude per layer (Figure 1b,d). In float64 autodiff tests at $d = 16$ across $\varepsilon \in \{10^{-8}, 10^{-5}, 10^{-2}\}$ and $r \in [10^{-2}, 10^{1}]$, the analytical formulas in Eq. (8) match numerical SVD values to maximum error 1.33×10^{-13}.

3.3 Antipodal Sign-Flip Instability and Tangent-Projected Geodesic SkipNorm

When building norm-preserving residual blocks on $\mathbb{R}^d$ ($d_{\text{in}} = d_{\text{out}} = d$), simply applying SkipNorm to a residual branch $y = x + \alpha f(x)$ yields **Projected Residual SkipNorm**:

$$\operatorname{ProjResSkipNorm}_\alpha(x; f) \coloneqq \frac{x + \alpha f(x)}{\|x + \alpha f(x)\|_2} \|x\|_2 . \tag{10}$$

While Eq. (10) preserves $\|\hat{y}\|_2 = \|x\|_2$, decompose $f(x) = f_r(x)u_{\text{in}} + f_\perp(x)$ into its radial component $f_r(x) = \langle f(x), u_{\text{in}} \rangle$ and tangential component $f_\perp(x) = (I_d - u_{\text{in}}u_{\text{in}}^\top)f(x) \in T_x\mathbb{S}^{d-1}(\|x\|_2)$. Whenever the radial component opposes x strongly enough that $1 + \alpha f_r(x)/\|x\|_2 < 0$ while $f_\perp(x) \to 0$, the numerator $x+\alpha f(x)$ crosses the origin and Eq. (10) discontinuously flips to the antipode $\hat{y} \to -x$ (Figure 1c). Projecting $f(x)$ onto $T_x\mathbb{S}^{d-1}(\|x\|_2)$ before spherical retraction defines **Geodesic SkipNorm**:

$$\text{GeodSkipNorm}_\alpha(x; f) \coloneqq \frac{x + \alpha f_\perp(x)}{\|x + \alpha f_\perp(x)\|_2} \|x\|_2, \qquad f_\perp(x) = f(x) - \frac{\langle f(x), x \rangle}{\|x\|_2^2} x. \tag{11}$$

Proposition 3 (Stability and First-Order Tangentiality of Geodesic SkipNorm)**.** *For all $x \in \mathbb{R}^d \setminus \{0\}$, all $\alpha \in \mathbb{R}$, and all vector fields $f : \mathbb{R}^d \to \mathbb{R}^d$:*

- ***(a) Strict Positivity and Antipodal Flip Elimination:*** *Because $\langle f_\perp(x), x \rangle = 0$, the denominator satisfies $\|x + \alpha f_\perp(x)\|_2 = \sqrt{\|x\|_2^2 + \alpha^2 \|f_\perp(x)\|_2^2} \geq \|x\|_2 > 0$, and the cosine similarity with the input is strictly positive:*

$$\frac{\langle \text{GeodSkipNorm}_\alpha(x; f), x \rangle}{\|x\|_2^2} = \frac{1}{\sqrt{1 + \alpha^2 \|f_\perp(x)\|_2^2 / \|x\|_2^2}} \in (0, 1]. \tag{12}$$

- ***(b) First-Order Tangential Retraction:*** *As $\alpha \to 0$, $\text{GeodSkipNorm}_\alpha(x; f) = x + \alpha f_\perp(x) - \frac{\alpha^2 \|f_\perp(x)\|_2^2}{2\|x\|_2^2} x + \mathcal{O}(\alpha^3)$, forming a second-order retraction onto $\mathbb{S}^{d-1}(\|x\|_2)$ that satisfies Proposition 1(b) ($J^\top u_{\text{out}} = u_{\text{in}}$) for every $\alpha \in \mathbb{R}$.*

4 Norm-Preserving Rollouts Without Post-Step Projection

Before investigating latent invariant manifolds in autoencoders, we evaluate Geodesic SkipNorm as a direct state-space propagator on a physical partial differential equation whose discrete L_2 norm is an exact conserved quantity. A critical methodological requirement is that the reference numerical solver must *never* apply post-step norm projection: any artificial renormalization in the training or test targets would conflate solver projection with model conservation.

Benchmark Protocol (1D Discrete Nonlinear Schrödinger Equation). We consider the focusing 1D periodic discrete nonlinear Schrödinger (DNLS) equation on an $M = 16$ lattice ($\psi_n(t) \in \mathbb{C}$, represented in real coordinates $x(t) = (\text{Re}\,\psi, \text{Im}\,\psi) \in \mathbb{R}^{32}$):

$$i\frac{d\psi_n}{dt} = -(\psi_{n+1} - 2\psi_n + \psi_{n-1}) - g_{\text{nl}} |\psi_n|^2 \psi_n, \qquad n \in \{1, \ldots, 16\}, \quad \psi_{n+16} = \psi_n, \tag{13}$$

with nonlinearity $g_{\text{nl}} = 1.0$, conserved total wavepacket power $\mathcal{N}(\psi) = \sum_{n=1}^{16} |\psi_n|^2 = \|x\|_2^2$, and conserved Hamiltonian $\mathcal{H}(\psi) = \sum_n (|\psi_{n+1} - \psi_n|^2 - \frac{g_{\text{nl}}}{2} |\psi_n|^4)$. We integrate 60 initial localized bright soliton wavepackets ($\mathcal{N}(\psi_0) \in [1.5, 4.5]$) over $T = 250$ macro-steps ($\Delta t = 0.04$, physical horizon $t_{\max} = 10.0$) using a 10×-substepped fourth-order Runge–Kutta solver ($\delta t = 0.004$) in float64 *with zero post-step normalization.* Across all 60 trajectories and 250 macro-steps, the intrinsic numerical norm drift of the unprojected RK4 reference solver is at most 1.32×10^{-11}. We partition the dataset strictly by whole trajectories into 36 training, 12 validation, and 12 held-out test orbits.

Results and Analysis. Table 1 and Figure 2 compare 7 architectures trained under an identical 5-step rollout MSE objective across 5 random seeds (2-layer MLP width 64, $\sim$ 6.3k parameters). Three findings stand out:

1. **Machine-Precision Conservation Without Projection:** Geodesic SkipNorm and Projected Residual SkipNorm preserve $\|\psi_t\|_2^2$ to 4.09×10^{-15} and 3.88×10^{-15} over 250 autoregressive steps—nearly four orders of magnitude tighter than the unprojected 10×-RK4 solver (1.32×10^{-11}). In

Table 1: **Five-seed 1D DNLS multi-step rollout benchmark ($d = 32$, 12 held-out test soliton wavepackets, 250 steps, no solver projection).** Models are trained with a 5-step rollout MSE loss across 5 random seeds (mean ± SD). Intrinsic unprojected 10×-RK4 reference solver norm drift is 1.32×10^{-11}.

Architecture	MSE@20 (↓)	MSE@50 (↓)	MSE@200 (↓)	Max Norm Drift (↓)	250-Step Time (ms)
Geodesic SkipNorm	0.00036 ± 0.00001	0.00257 ± 0.00014	0.06417 ± 0.00236	$\mathbf{4.09 \times 10^{-15}}$	3.74 ± 0.34
Projected Residual SkipNorm	0.00036 ± 0.00001	0.00254 ± 0.00014	0.06340 ± 0.00238	$\mathbf{3.88 \times 10^{-15}}$	3.94 ± 0.31
State-Dependent SO(32) (Cayley)	$\mathbf{0.00021 \pm 0.00002}$	$\mathbf{0.00134 \pm 0.00016}$	0.03546 ± 0.00096	$\mathbf{3.08 \times 10^{-15}}$	27.55 ± 0.90
Linear Orthogonal SO(32)	0.00043 ± 0.00001	0.00241 ± 0.00004	$\mathbf{0.03485 \pm 0.00026}$	2.22×10^{-14}	$\mathbf{2.73 \pm 0.17}$
Pre-RMSNorm Residual	0.00087 ± 0.00007	0.00600 ± 0.00041	0.29810 ± 0.03768	39.20 ± 5.18	3.80 ± 0.11
Post-RMSNorm Residual	0.57798 ± 0.00011	0.58874 ± 0.00104	0.79204 ± 0.00780	17.41 ± 0.00	3.92 ± 0.26
Unconstrained ResNet	0.00040 ± 0.00002	0.00292 ± 0.00027	1.29241 ± 0.45157	387.04 ± 130.52	3.69 ± 0.13

contrast, Post-RMSNorm resets $\|\psi_t\|_2 = \sqrt{32}$ at step 1 (yielding 17.4× relative norm error on low-energy wavepackets), while Pre-RMSNorm Residual and Unconstrained ResNet accumulate massive norm drift (39.20 and 387.04) and diverge at long horizons (MSE@200 = 0.298 and 1.292 vs. 0.0642 for Geodesic SkipNorm).

2. **Efficiency vs. State-Dependent** SO(32) **Cayley Maps:** Parameterizing a state-dependent orthogonal matrix $Q(x) = (I - S(x))(I + S(x))^{-1} \in \mathrm{SO}(32)$ requires solving a 32×32 linear system at every step, increasing 250-step rollout latency by 7.4× (27.55 ms vs. 3.74 ms for Geodesic SkipNorm, which operates in $\mathcal{O}(d)$ vector arithmetic).

3. **Accuracy and Expressivity:** Geodesic SkipNorm achieves lower medium- and long-horizon error than both Unconstrained ResNet (MSE@50 = 0.00257 vs. 0.00292) and Pre-RMSNorm Residual (0.00600), and outperforms Linear Orthogonal SO(32) at short horizons (MSE@20 = 0.00036 vs. 0.00043) while retaining full nonlinear expressivity for strongly non-isometric coordinate systems (Sections 5–7).

5 Conditional Radial Foliation of Conserved Quantities

In general conservative systems (such as Lotka–Volterra population dynamics or anharmonic oscillators), the conserved quantity $I(x)$ is *not* the Euclidean norm $\|x\|_2$ in observed coordinates. Consider a dynamical autoencoder (E, g, D) with encoder $E : \mathcal{M} \to \mathbb{R}^{d_z}$, norm-preserving latent transition $g : \mathbb{R}^{d_z} \to \mathbb{R}^{d_z}$ ($\|g(z)\|_2 = \|z\|_2$, implemented via Geodesic SkipNorm or $\mathrm{SO}(d_z)$), and decoder $D : \mathbb{R}^{d_z} \to \mathcal{M}$. We now state and prove the exact conditions under which intertwining the observed flow $\Phi_{\Delta t}$ with g forces the latent radius $h(x) \coloneqq \|E(x)\|_2$ to foliate the level sets of I.

5.1 The Conditional Radial Foliation Theorem

Theorem 1 (Conditional Radial Foliation of Connected Regular Level Sets)**.** *Let $U \subseteq \mathbb{R}^N$ ($N \geq 2$) be an open domain and let $I \in C^1(U, \mathbb{R})$ be a conserved quantity of a discrete-time flow map $\Phi_{\Delta t} : U \to U$ ($I \circ \Phi_{\Delta t} = I$). For $c \in (c_{\min}, c_{\max})$, denote the level set by $\mathcal{O}_c \coloneqq \{x \in U : I(x) = c\}$. Suppose an encoder $E : U \to \mathbb{R}^{d_z}$ ($d_z \geq 2$) and a latent map $g : \mathbb{R}^{d_z} \to \mathbb{R}^{d_z}$ satisfy:*

- ***(H1) Exact Intertwining with a Norm-Preserving Map:*** *E is continuous on U, $\|g(z)\|_2 = \|z\|_2$ for all $z \in E(U)$, and $E \circ \Phi_{\Delta t} = g \circ E$ on U.*
- ***(H2) Connectedness and Topological Orbit Density:*** *For each $c \in (c_{\min}, c_{\max})$, $\mathcal{O}_c$ is connected and there exists $x_0 \in \mathcal{O}_c$ whose forward orbit $\{\Phi_{\Delta t}^k(x_0)\}_{k \in \mathbb{N}}$ is dense in $\mathcal{O}_c$.*

Then:

- ***(a) Level-Set Radial Constancy (Arbitrary $d_z \geq 2$):*** *The latent radius $h(x) \coloneqq \|E(x)\|_2$ is constant along every orbit, and by continuity and (H2) there exists a unique function $\psi : (c_{\min}, c_{\max}) \to \mathbb{R}_{\geq 0}$*

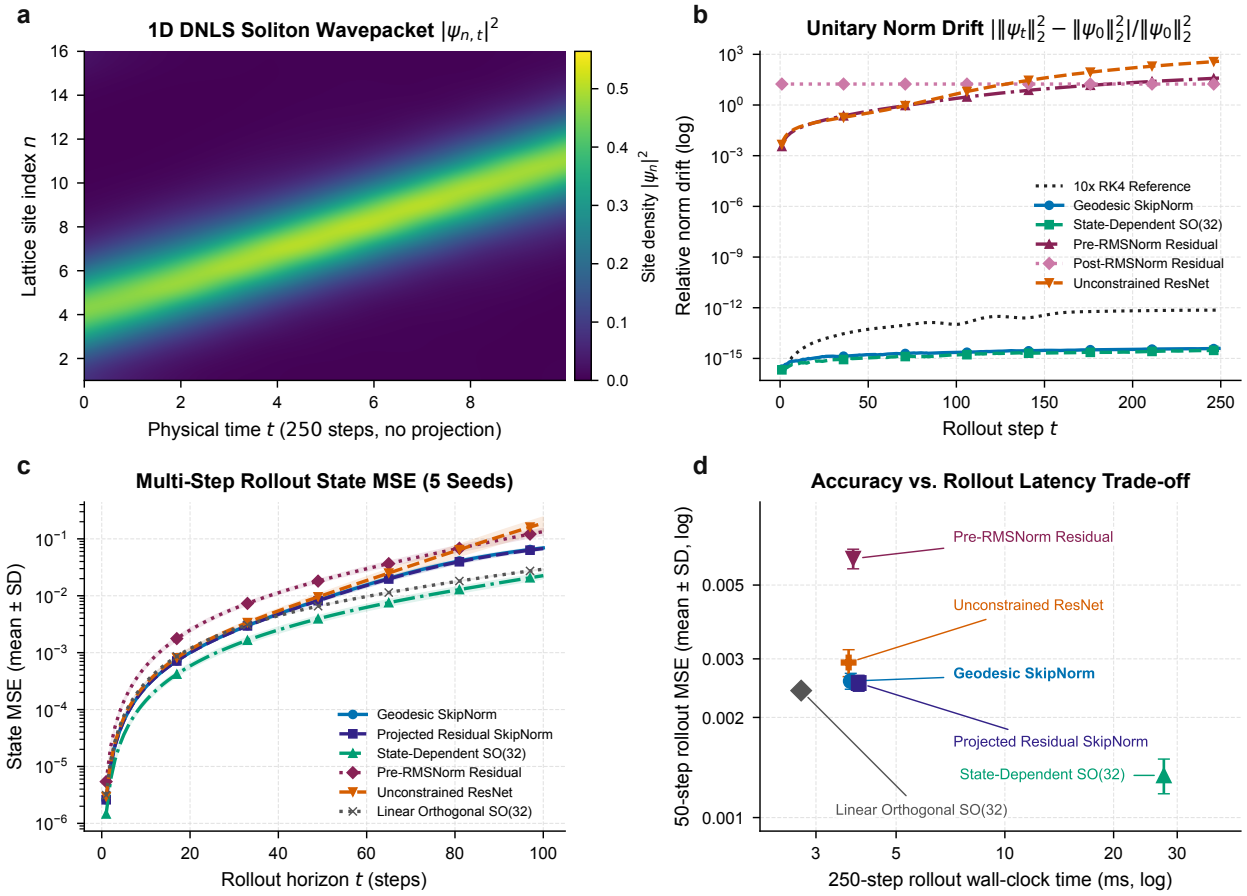


Figure 2: **Unitary soliton wavepacket rollout on the focusing 1D discrete nonlinear Schrödinger (DNLS) equation ($d = 32$) without reference solver projection.** **(a)** Spatiotemporal site density $|\psi_{n,t}|^2$ of a held-out bright soliton wavepacket propagating across the lattice over 250 steps. **(b)** Relative wavepacket power drift $\left|\|\psi_t\|_2^2 - \|\psi_0\|_2^2\right| / \|\psi_0\|_2^2$: Geodesic SkipNorm and SO(32) transitions maintain machine-precision conservation (~ 3–4×10^{-15}), below even the unprojected 10×-RK4 reference solver drift (1.32×10^{-11}), whereas Pre-RMSNorm and Unconstrained ResNet drift by $\mathcal{O}(10^1$–$10^2)$ and Post-RMSNorm collapses immediately onto the fixed $\sqrt{d}$ sphere. **(c)** Held-out state MSE versus rollout horizon across 5 random seeds (mean ± SD). **(d)** Accuracy (MSE@50) versus 250-step rollout wall-clock time: Geodesic SkipNorm outperforms Unconstrained ResNet and Pre-RMSNorm Residual while running 7.4× faster than state-dependent SO(32) Cayley steps.

such that

$$\|E(x)\|_2 = \psi(I(x)) \quad \forall x \in \bigcup_{c \in (c_{\min}, c_{\max})} \mathcal{O}_c, \qquad E(\mathcal{O}_c) \subseteq \mathbb{S}^{d_z - 1}(\psi(c)). \tag{14}$$

*If, in addition, every $c \in (c_{\min}, c_{\max})$ is a **regular value** of I ($\nabla I(x) \neq 0$ on $\mathcal{O}_c$), then I admits continuous local sections and $\psi : (c_{\min}, c_{\max}) \to \mathbb{R}_{\geq 0}$ is continuous.*

(b) Dimension-Matched Concentric Homeomorphism and Strict Monotonicity *($d_z = N$): Assume further:*

(H3) *$d_z = N \geq 2$, every $c \in (c_{\min}, c_{\max})$ is a regular value of I with $\mathcal{O}_c$ compact and connected, and $E : U \to \mathbb{R}^N$ is globally **injective** on U.*

*Then $\psi(c) > 0$ for all $c \in (c_{\min}, c_{\max})$, the restriction $E|_{\mathcal{O}_c} : \mathcal{O}_c \to \mathbb{S}^{N-1}(\psi(c))$ is a **homeomorphism onto the entire sphere** $\mathbb{S}^{N-1}(\psi(c))$, and $\psi : (c_{\min}, c_{\max}) \to \mathbb{R}_{>0}$ is a continuous, **strictly monotone** bijection onto an open interval $(\psi_{\min}, \psi_{\max}) \subseteq \mathbb{R}_{>0}$.*

The complete proof is given in Appendix B. Briefly, part (a) follows from $\left\|E(\Phi^k_{\Delta t}(x_0))\right\|_2 = \left\|g^k(E(x_0))\right\|_2 = \|E(x_0)\|_2$ and continuity of $\|E(\cdot)\|_2$ on the closure $\overline{\{\Phi^k_{\Delta t}(x_0)\}} = \mathcal{O}_c$, while continuity of ψ follows from the Implicit Function Theorem local section $\sigma : (c-\delta, c+\delta) \to U$ $(I(\sigma(s)) = s \implies \psi(s) = \|E(\sigma(s))\|_2)$. In part (b), each regular compact connected level set $\mathcal{O}_c$ is a closed $(N-1)$-manifold and $\mathbb{S}^{N-1}(\psi(c))$ is a connected $(N-1)$-manifold $(N \geq 2)$; by Brouwer's Invariance of Domain (Brouwer, 1911), the continuous injective map $E|_{\mathcal{O}_c}$ has an image that is both open (invariance of domain) and closed (compactness of $\mathcal{O}_c$) in the connected sphere $\mathbb{S}^{N-1}(\psi(c))$, forcing $E(\mathcal{O}_c) = \mathbb{S}^{N-1}(\psi(c))$. Global injectivity of E across distinct level sets $c_1 \neq c_2$ then guarantees $\mathbb{S}^{N-1}(\psi(c_1)) \cap \mathbb{S}^{N-1}(\psi(c_2)) = \emptyset \implies \psi(c_1) \neq \psi(c_2)$, and a continuous injective real function on an interval is strictly monotone.

Why Topological Orbit Density (H2) is Essential: The Resonant $1:1$ Oscillator Control. Suppose $N = 4$ and a 4D phase space has a single scalar energy $H(q_1, p_1, q_2, p_2) = \frac{1}{2}(q_1^2 + p_1^2) + \frac{1}{2}(q_2^2 + p_2^2) = H_1 + H_2$, so each energy shell $\mathcal{O}_c \cong \mathbb{S}^3(\sqrt{2c})$ is compact, connected, and regular, *but* every orbit is periodic ($1:1$ resonance) and therefore a 1D circle $\mathbb{S}^1$ that is **nowhere dense** in $\mathbb{S}^3(\sqrt{2c})$. Consider the analytical diffeomorphism $E(q_1, p_1, q_2, p_2) = (\sqrt{2}q_1, \sqrt{2}p_1, q_2, p_2)$ intertwined with the exact block-rotation $g \in \mathrm{SO}(4)$. As verified in Figure 4c and Appendix B:

$$E \circ \Phi_{\Delta t} = g \circ E \ \text{(exact)}, \qquad \mathrm{CV}_{\mathcal{O}}(h) = 4.17 \times 10^{-16} \ \text{(zero within-orbit wobble)}, \tag{15}$$

yet on the single energy shell $H = 1$, the encoded radius $\|E(x)\|_2 = \sqrt{2H_1 + 2H} = \sqrt{2(1+\eta)}$ varies continuously with the mode energy partition $\eta = H_1/H \in [0,1]$ from $\sqrt{2} \approx 1.414$ to 2.000 (a 50% relative spread across orbits on the same level set). Thus, without (H2), intertwining only forces $\|E(x)\|_2$ to be an orbit invariant, not a function of H alone.

5.2 Evaluation Metrics and Whole-Orbit Split Protocol

To evaluate empirical approximations of Theorem 1 without conflating transition preservation with encoder radial constancy, all radial foliation metrics are evaluated **strictly on encoded ground-truth test trajectories** $z_{m,t} = E(x_{m,t})$ $(m = 1, \ldots, M_{\text{test}}, t = 0, \ldots, T-1)$ rather than on latent rollouts $g^t(E(x_{m,0}))$ (which trivially have constant norm whenever g is norm-preserving). Defining $r_{m,t} \coloneqq \|E(x_{m,t})\|_2$, the per-orbit mean radius $\bar{r}_m \coloneqq \frac{1}{T}\sum_{t=0}^{T-1} r_{m,t}$, and within-orbit standard deviation $s_m \coloneqq \sqrt{\frac{1}{T}\sum_{t=0}^{T-1}(r_{m,t} - \bar{r}_m)^2}$, we report:

1. **Within-Orbit Radial Coefficient of Variation ($\mathrm{CV}_{\mathcal{O}}(h)$):** $\mathrm{CV}_{\mathcal{O}}(h) \coloneqq \frac{1}{M_{\text{test}}}\sum_{m=1}^{M_{\text{test}}} \frac{s_m}{\bar{r}_m + \epsilon}$.

2. **Degenerate-Denominator-Guarded Orbit Variance Ratio (OVR):** Let $\sigma^2_{\text{lat}} = \frac{1}{M_{\text{test}}T}\sum_{m,t}\|z_{m,t} - \bar{z}\|_2^2$ be the total latent variance. To prevent scale collapse ($\mathrm{Var}_m(\bar{r}_m) \to 0$) from masquerading as small ratio,

$$\mathrm{OVR} \coloneqq \begin{cases} \infty & \text{if } \mathrm{Var}_m(\bar{r}_m) < 10^{-4}\sigma^2_{\text{lat}}, \\ \dfrac{\frac{1}{M_{\text{test}}}\sum_{m=1}^{M_{\text{test}}} s_m^2}{\mathrm{Var}_m(\bar{r}_m)} & \text{otherwise.} \end{cases} \tag{16}$$

3. **Monotone Foliation Fit (Isotonic R^2 and Spearman $|\rho_S|$):** The out-of-sample R^2 of a 1D monotone (increasing or decreasing) isotonic regression fitted on $(\{c_j^{\text{train}}\}, \{\bar{r}_j^{\text{train}}\})$ and evaluated on held-out test orbits $(\{c_m^{\text{test}}\}, \{\bar{r}_m^{\text{test}}\})$, alongside the rank correlation $|\rho_S(\{\bar{r}_m^{\text{test}}\}, \{c_m^{\text{test}}\})|$.

Table 2: **Five-seed Lotka–Volterra scalar invariant foliation benchmark (16 held-out whole test orbits, $T = 180$).** All metrics are evaluated on encoded ground-truth test trajectories $z_t = E(x_t)$ across 5 random seeds (mean ± SD).

Model ($d_z = 2$)	**Within-Orbit** $\mathrm{CV}_{\mathcal{O}}(h)$ (↓)	**Guarded** OVR (↓)	**Isotonic** R^2 (↑)	**Spearman** $\lvert\rho_S\rvert$ (↑)	**Rollout MSE** (↓)
Geodesic SkipNorm (MLP)	0.0265 ± 0.0027	0.0049 ± 0.0012	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	0.0028 ± 0.0009
Geodesic SkipNorm (Bijective)	$\mathbf{0.0135 \pm 0.0046}$	0.0019 ± 0.0012	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	0.0010 ± 0.0004
Projected Residual SkipNorm	0.0148 ± 0.0008	0.0017 ± 0.0002	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	0.0003 ± 0.0001
State-Dependent SO(2)	0.0141 ± 0.0011	$\mathbf{0.0015 \pm 0.0002}$	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{0.0002 \pm 0.0000}$
COMET Baseline (Otto et al., 2023)	0.3239 ± 0.1658	0.4949 ± 0.2501	$\mathbf{1.0000 \pm 0.0000}$	0.9994 ± 0.0012	0.0003 ± 0.0002
Unconstrained Autoencoder	0.1539 ± 0.0061	0.2196 ± 0.0106	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{0.0002 \pm 0.0000}$

Lotka–Volterra 5-Seed Foliation Results. We test Theorem 1 on Lotka–Volterra predator–prey dynamics in log-coordinates $(u, v) = (\log x_1, \log x_2) \in \mathbb{R}^2$, $\dot{u} = 1 - e^v$, $\dot{v} = e^u - 1$, whose conserved Hamiltonian $I(u, v) = e^u - u + e^v - v - 2 > 0$ has compact, connected, non-circular level sets surrounding the equilibrium $(0, 0)$ (Figure 3a). Using 80 full closed orbits ($T = 180$, $\Delta t = 0.05$) split into 48 train, 16 validation, and 16 held-out test orbits, Table 2 and Figure 3 confirm that all four norm-preserving latent transitions (Geodesic SkipNorm with MLP or bijective RealNVP encoder, Projected Residual SkipNorm, and State-Dependent SO(2)) map the non-circular observed contours onto concentric circles in $\mathbb{R}^2$ (Figure 3b), achieving $\mathrm{CV}_{\mathcal{O}}(h) \in [0.0135, 0.0265]$, guarded OVR $\in [0.0015, 0.0049]$, and held-out isotonic $R^2 = 1.0000$ ($|\rho_S| = 1.0000$). Crucially, while an Unconstrained Autoencoder preserves coarse orbit ordering ($|\rho_S| = 1.0000$), its encoded trajectories remain ovular with $5.8\times$ –$11.4\times$ larger within-orbit radial wobble ($\mathrm{CV}_{\mathcal{O}}(h) = 0.1539$, OVR $= 0.2196$; Figure 3c).

6 Topological Obstructions on Multi-Basin Phase Spaces

What happens when the connectedness hypothesis (H2) of Theorem 1 is violated, as in a double-well Hamiltonian $H(q, p) = \frac{1}{2}p^2 - \frac{1}{2}q^2 + \frac{1}{4}q^4 + \gamma q$ at sub-barrier energies $c < c_{\text{saddle}}$ where each level set $\mathcal{O}_c = \mathcal{O}_c^{(1)} \sqcup \mathcal{O}_c^{(2)}$ is the disjoint union of a left-well loop ($q < q_{\text{saddle}}$) and a right-well loop ($q > q_{\text{saddle}}$)?

Theorem 2 (Jordan–Brouwer Topological Obstruction on Disconnected Level Sets)**.** *Let $U \subseteq \mathbb{R}^N$ ($N \geq 2$) be an open domain containing two disjoint compact connected components $\mathcal{O}_{c_1}^{(1)}$ and $\mathcal{O}_{c_2}^{(2)}$, each homeomorphic to $\mathbb{S}^{N-1}$, such that their bounded Jordan–Brouwer interiors $\mathrm{Int}(\mathcal{O}_{c_1}^{(1)})$ and $\mathrm{Int}(\mathcal{O}_{c_2}^{(2)})$ in $\mathbb{R}^N$ are disjoint and contained in U. Let $E : U \to \mathbb{R}^N$ be any **continuous injective** map.*

- ***(a) No Simultaneous Origin-Enclosing Embedding:*** *By the Jordan–Brouwer Separation Theorem and Brouwer's Invariance of Domain, $E(\mathrm{Int}(\mathcal{O}_{c_1}^{(1)})) = \mathrm{Int}(E(\mathcal{O}_{c_1}^{(1)}))$ and $E(\mathrm{Int}(\mathcal{O}_{c_2}^{(2)})) = \mathrm{Int}(E(\mathcal{O}_{c_2}^{(2)}))$ are disjoint open topological balls in $\mathbb{R}^N$. Consequently, the latent origin $0 \in \mathbb{R}^N$ **cannot lie in the bounded interior of both** $E(\mathcal{O}_{c_1}^{(1)})$ and $E(\mathcal{O}_{c_2}^{(2)})$.*

- ***(b) No Concentric Spherical Foliation:*** *No continuous injective encoder $E : U \to \mathbb{R}^N$ can map both $\mathcal{O}_{c_1}^{(1)}$ and $\mathcal{O}_{c_2}^{(2)}$ onto spheres centered at the origin ($\mathbb{S}^{N-1}(r_1)$ and $\mathbb{S}^{N-1}(r_2)$, whether $r_1 = r_2$ or $r_1 \neq r_2$), because every origin-centered sphere $\mathbb{S}^{N-1}(r)$ ($r > 0$) has 0 in its bounded interior $B_N(0, r)$.*

- ***(c) Consequences for Trained Autoencoders:*** *Any autoencoder (E, g, D) on a multi-well phase space must exhibit at least one of three behaviors:*

  (i) Origin-Pinned Bijective Regime: *If E is globally injective and prevented from large off-origin translation, at least one well cannot enclose $0 \in \mathbb{R}^N$ and therefore cannot lie on an origin-centered sphere, forcing large within-orbit radial variation $\mathrm{CV}_{\mathcal{O}}(h) > 0$.*

  (ii) Non-Injective Well-Folding Regime: *If E is an unconstrained MLP, it can fold both wells onto concentric origin-enclosing spheres ($\|E(x)\|_2 \approx \psi(H(x))$), achieving small $\mathrm{CV}_{\mathcal{O}}(h)$ at the cost of violating global injectivity ($E(\mathcal{O}_c^{(1)}) \cap E(\mathcal{O}_c^{(2)}) \neq \emptyset$) and collapsing cross-well separability.*

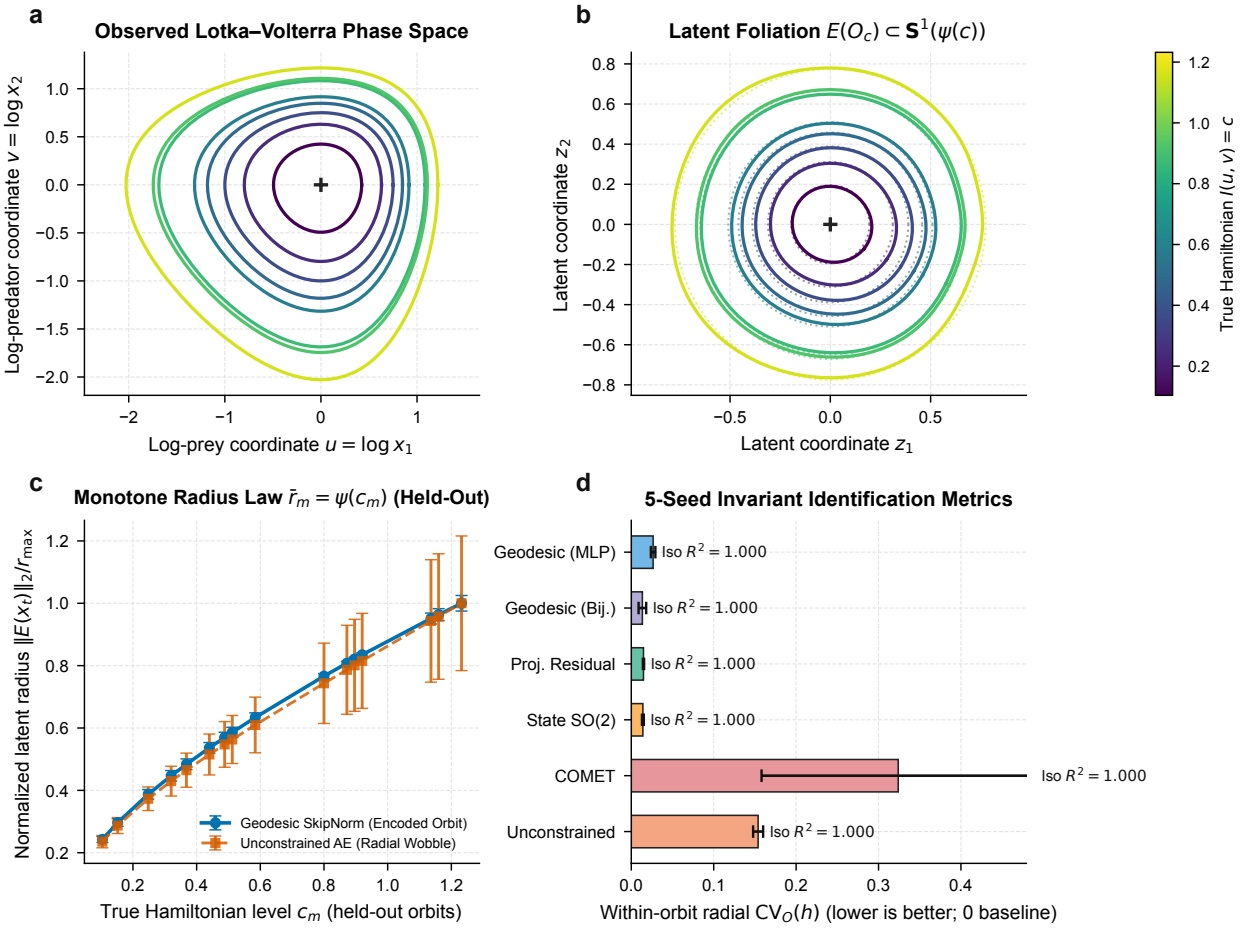


Figure 3: **Radial foliation of the conserved Lotka–Volterra Hamiltonian on held-out whole test orbits (Theorem 1). (a)** Observed test trajectories in log-population phase space $(u, v) = (\log x_1, \log x_2)$, where level sets of $I(u, v) = e^u - u + e^v - v - 2$ are strongly non-circular closed ovular contours. **(b)** Encoded test trajectories $z_t = E(x_t)$ under Geodesic SkipNorm ($d_z = 2$), mapping each closed orbit onto a concentric circle $\mathbb{S}^1(\psi(c))$ (dotted reference circles). **(c)** Normalized mean latent radius $\bar{r}_m / r_{\max}$ versus true Hamiltonian c_m on the 16 held-out test orbits (± 1 SD error bars reflect within-orbit radial fluctuation s_m): Geodesic SkipNorm achieves $\mathrm{CV}_{\mathcal{O}}(h) = 0.0135$–$0.0265$ and isotonic $R^2 = 1.0000$, whereas the Unconstrained AE exhibits $5.8\times$ –$11.4\times$ larger within-orbit radial wobble ($\mathrm{CV}_{\mathcal{O}}(h) = 0.1539$). **(d)** Five-seed comparison of within-orbit radial $\mathrm{CV}_{\mathcal{O}}(h)$ and held-out isotonic R^2.

(iii) Off-Origin Translation Bypass: *An injective map $E_a(x) = ae_1 + \tilde{E}(x)$ shifted by $\|ae_1\|_2 = a \gg R$ places both wells outside 0 on a tiny patch of $\mathbb{S}^{N-1}(a)$, achieving $\mathrm{CV}_{\mathcal{O}}(h) = \mathcal{O}(a^{-1})$ and relative intertwining error $\mathcal{O}(a^{-1})$ trivially by collapsing relative radius variation across the dataset.*

Table 3 and Figure 4a,b,d verify all three regimes of Theorem 2 across 5 seeds and three sub-barrier configurations (Balanced 50:50, Imbalanced 80:20, and Tilted $\gamma = 0.08$ where left/right wells occupy distinct energy intervals). Across all $3\times2\times5 = 30$ bijective RealNVP runs (`Origin-Pinned` and `Free-Shift`), the fraction of runs in which both wells enclose $0 \in \mathbb{R}^2$ is **identically** 0.00 (Figure 4a), forcing $\mathrm{CV}_{\mathcal{O}}(h) \in [0.237, 0.281]$ when origin-pinned. Conversely, the unconstrained `Well-Folding MLP` folds both wells around 0 (`BothEnclose0` = 0.80–1.00), reducing $\mathrm{CV}_{\mathcal{O}}(h)$ to 0.081–0.100 (Figure 4b) by sacrificing global injectivity (`Well-Sep Acc` drops from 0.850 to 0.625). Finally, Figure 4d verifies the analytical translated-encoder control $E_a(x) = ae_1 + x$ with identity latent map $g(z) = z$ (Appendix C): as a increases from 2 to 10^4, $\mathrm{CV}_{\mathcal{O}}(h)$ decays

Table 3: **Five-seed Double-Well Jordan–Brouwer obstruction benchmark across three sub-barrier regimes (Theorem 2).** "BothEnclose0" denotes the fraction of seeds in which both left- and right-well encoded test loops enclose $0 \in \mathbb{R}^2$; "Well-Sep Acc" is linear probe accuracy separating left vs. right wells from latent radius $\bar{r}_m$.

Sub-Barrier Regime	Encoder Class ($d_z = 2$)	BothEnclose0	Radial $\mathrm{CV}_{\mathcal{O}}(h)$	Within-Well R^2	Global R^2	Well-Sep Acc
Balanced (50:50, $\gamma = 0$)	Origin-Pinned Bijective	**0.00**	0.2814 ± 0.0472	1.0000 ± 0.0000	0.9163 ± 0.0725	0.750 ± 0.056
	Well-Folding MLP	1.00	$\mathbf{0.0812 \pm 0.0253}$	1.0000 ± 0.0000	$\mathbf{0.9710 \pm 0.0219}$	0.625 ± 0.040
	Free-Shift Bijective	**0.00**	0.1183 ± 0.0365	0.9613 ± 0.0774	0.7145 ± 0.1597	$\mathbf{0.850 \pm 0.146}$
Imbalanced (80:20, $\gamma = 0$)	Origin-Pinned Bijective	**0.00**	0.2446 ± 0.0439	0.9985 ± 0.0029	0.8229 ± 0.1059	0.812 ± 0.079
	Well-Folding MLP	0.80	$\mathbf{0.1003 \pm 0.0258}$	1.0000 ± 0.0000	$\mathbf{0.8888 \pm 0.1631}$	0.688 ± 0.112
	Free-Shift Bijective	**0.00**	0.1257 ± 0.0221	0.9981 ± 0.0038	0.6606 ± 0.0879	$\mathbf{0.875 \pm 0.079}$
Tilted ($\gamma = 0.08$)	Origin-Pinned Bijective	**0.00**	0.2369 ± 0.0209	0.9999 ± 0.0003	0.8962 ± 0.0528	0.850 ± 0.075
	Well-Folding MLP	1.00	$\mathbf{0.0846 \pm 0.0472}$	0.9991 ± 0.0017	$\mathbf{0.9614 \pm 0.0228}$	0.713 ± 0.073
	Free-Shift Bijective	**0.00**	0.1262 ± 0.0268	0.9265 ± 0.1456	0.7938 ± 0.0725	$\mathbf{0.875 \pm 0.056}$

as $\mathcal{O}(a^{-1})$ down to 2.10×10^{-5}, confirming why the guarded OVR metric in Eq. (16) and origin-centering constraints are essential.

7 Toroidal Generalization: K-Block SkipNorm on $\bigoplus_{k=1}^{K} \mathbb{R}^{d_k}$

An integrable system with $K \geq 2$ independent commuting invariants $I = (I_1, \ldots, I_K) : U \to \mathbb{R}^K$ has compact regular joint level sets $\mathcal{O}_c = \{x \in U : I(x) = c\}$ diffeomorphic to the K-torus $\mathbb{T}^K = (\mathbb{S}^1)^K$ by the Liouville–Arnold theorem (Arnold, 1989). A single spherical SkipNorm layer preserves only the total Euclidean norm $\|z\|_2 = \sqrt{\sum_{k=1}^{K} \left\|z^{(k)}\right\|_2^2}$, which projects the K-dimensional invariant vector $c \in \mathbb{R}^K$ onto a 1D scalar radius and cannot separate $\mathbb{T}^K$ foliations. To match the toroidal topology $\mathbb{T}^K$, we partition the latent space into K orthogonal blocks $z = (z^{(1)}, \ldots, z^{(K)}) \in \bigoplus_{k=1}^{K} \mathbb{R}^{d_k}$ ($d_k \geq 2$) and apply **K-Block Toroidal Geodesic SkipNorm** independently to each block:

$$g_{\text{block}}(z) \coloneqq \Big(\text{GeodSkipNorm}_{\alpha_1}\big(z^{(1)}; f^{(1)}(z)\big), \ \ldots, \ \text{GeodSkipNorm}_{\alpha_K}\big(z^{(K)}; f^{(K)}(z)\big) \Big), \tag{17}$$

where each branch $f^{(k)}(z)$ depends on the full latent state z (capturing nonlinear inter-mode frequency coupling), while its spherical retraction preserves each block radius $\rho_k(z) \coloneqq \left\|z^{(k)}\right\|_2$ individually: $\rho_k(g_{\text{block}}(z)) = \rho_k(z)$ for all $k \in \{1, \ldots, K\}$.

Theorem 3 (Single-Sphere Non-Identifiability and K-Block Action Recovery on $\mathbb{T}^K$)**.** *Let $I = (I_1, \ldots, I_K) \in C^1(U, \mathbb{R}^K)$ ($K \geq 2$) be K functionally independent invariants of $\Phi_{\Delta t} : U \to U$ whose regular level sets $\mathcal{O}_c = I^{-1}(c)$ ($c \in \Omega \subseteq \mathbb{R}^K$ open) are compact, connected K-tori $\mathbb{T}^K$ carrying dense quasi-periodic orbits.*

- ***(a) Single-Sphere Information Bottleneck ($K = 1$):*** *Under any continuous encoder $E : U \to \mathbb{R}^{d_z}$ intertwined with a single-sphere norm-preserving map ($\|g(z)\|_2 = \|z\|_2$), the total radius $\|E(x)\|_2 = \psi(I(x))$ is a scalar function of $c \in \Omega \subseteq \mathbb{R}^K$. By the Rank Theorem, for every regular value of ψ, the preimage $\psi^{-1}(r) \subset \Omega$ is a $(K-1)$-dimensional submanifold of distinct action vectors $c \neq c'$ assigned the **identical** latent radius r, so no readout from $\|E(x)\|_2$ can recover $c \in \mathbb{R}^K$. Furthermore, since $\pi_1(\mathbb{T}^K) \cong \mathbb{Z}^K$ whereas $\pi_1(\mathbb{S}^{N-1}) = 0$ for $N \geq 3$, $\mathbb{T}^K$ cannot embed homeomorphically onto $\mathbb{S}^{N-1}$.*
- ***(b) K-Block Toroidal Foliation on $\bigoplus_{k=1}^{K} \mathbb{R}^2$:*** *Suppose $E = (E^{(1)}, \ldots, E^{(K)}) : U \to \bigoplus_{k=1}^{K} \mathbb{R}^2$ intertwines $\Phi_{\Delta t}$ with a K-block norm-preserving map g_{block} ($\left\|g_{\text{block}}^{(k)}(z)\right\|_2 = \left\|z^{(k)}\right\|_2$). By Theorem 1(a) applied to each block norm $\rho_k(E(x)) = \left\|E^{(k)}(x)\right\|_2$, there exists a continuous map $\Psi = (\psi_1, \ldots, \psi_K) : \Omega \to \mathbb{R}^K_{\geq 0}$ such that*

$$\boldsymbol{\rho}(E(x)) \coloneqq \Big(\left\|E^{(1)}(x)\right\|_2, \ldots, \left\|E^{(K)}(x)\right\|_2 \Big) = \Psi(I(x)) \quad \forall x \in \bigcup_{c \in \Omega} \mathcal{O}_c, \qquad E(\mathcal{O}_c) \subseteq \prod_{k=1}^{K} \mathbb{S}^1(\psi_k(c)). \tag{18}$$

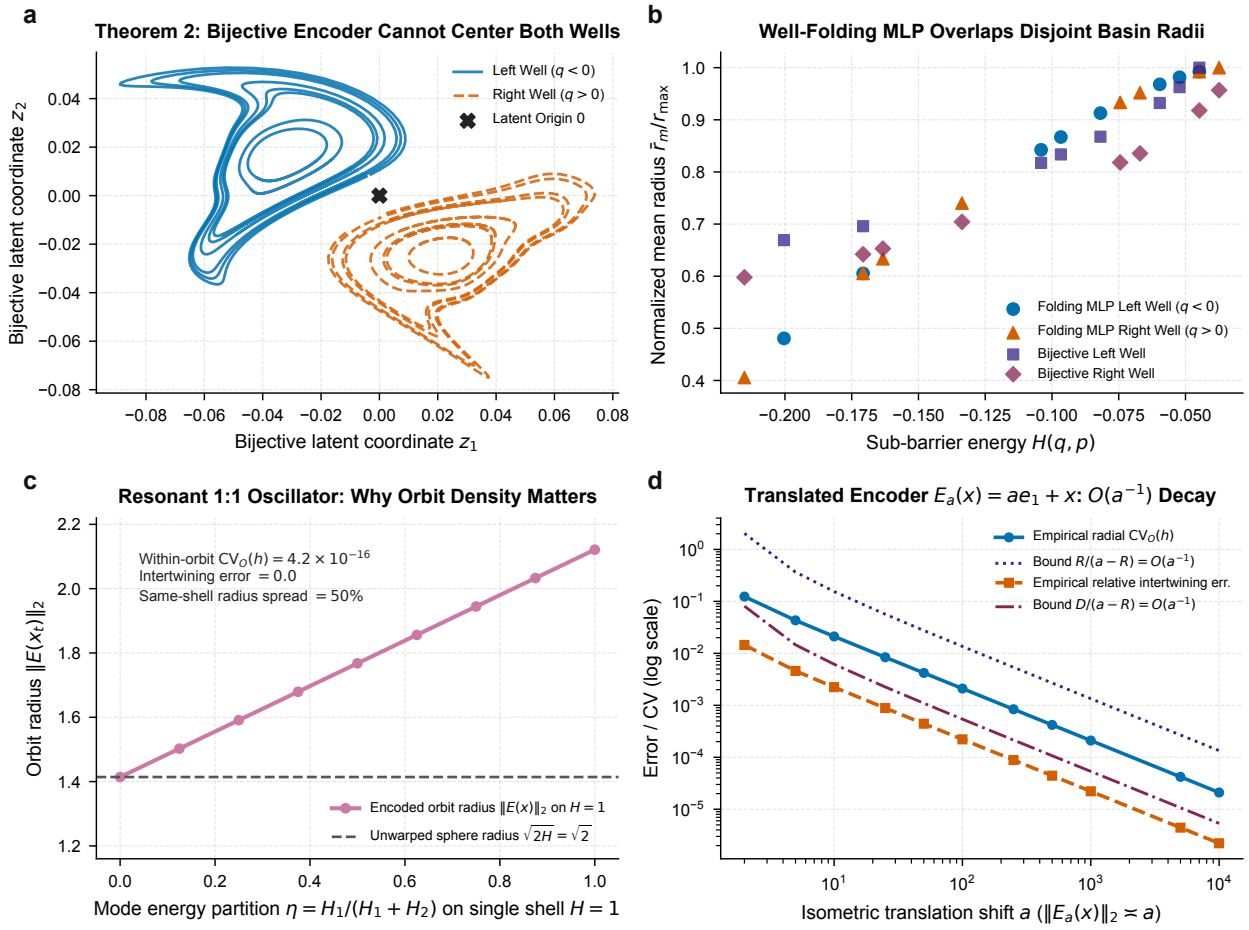


Figure 4: **Topological obstructions and analytical controls (Theorem 1 and Theorem 2). (a)** Bijective RealNVP encoder on sub-barrier Double-Well orbits: exactly as proved by Theorem 2(a), the disjoint left-well (blue solid) and right-well (orange dashed) image loops cannot both enclose the latent origin $0 \in \mathbb{R}^2$ (`BothEnclose0 = 0.00` across all 15 bijective runs). **(b)** Normalized mean radius $\bar{r}_m/r_{\max}$ versus sub-barrier energy $H(q, p)$: the unconstrained Well-Folding MLP folds both wells around 0 so their radii overlap at identical energies, whereas the bijective encoder separates the two basins at the cost of non-circular off-origin distortion. **(c)** Resonant $1:1$ 2D harmonic oscillator control in $\mathbb{R}^4$: exact SO(4) intertwining yields $\mathrm{CV}_{\mathcal{O}}(h) = 4.2 \times 10^{-16}$, yet lack of topological orbit density on $H = 1$ produces a 50% radius spread across mode partitions $\eta = H_1/(H_1 + H_2)$. **(d)** Translated identity encoder control $E_a(x) = ae_1 + x, g(z) = z$: both within-orbit $\mathrm{CV}_{\mathcal{O}}(h)$ and relative intertwining error decay as $\mathcal{O}(a^{-1})$ strictly by moving the dataset off the origin.

> *If E is an injective immersion on U with $\psi_k(c) > 0$, then $E|_{\mathcal{O}_c} : \mathcal{O}_c \to \prod_{k=1}^{K} \mathbb{S}^1(\psi_k(c)) \cong \mathbb{T}^K$ is a homeomorphism onto the poly-circle and $\Psi : \Omega \to \Psi(\Omega) \subseteq \mathbb{R}^K_{>0}$ is a continuous injective coordinate chart that uniquely identifies the full invariant vector $c = (I_1(x), \ldots, I_K(x))$ up to a C^0 homeomorphism on $\mathbb{R}^K$.*

4D Coupled Nonlinear Oscillator Benchmark ($K = 2$). We test Theorem 3 on a 4D system with two hidden action invariants $(I_1, I_2) \in [0.3, 1.5]^2$ and incommensurate nonlinear frequencies $\omega_1(I) = 1.0 + 0.35I_1 + 0.15I_2$, $\omega_2(I) = \sqrt{2} + 0.20I_1 + 0.40I_2$, observed through a smooth nonlinear diffeomorphism $x = \Phi_{\mathrm{mix}}(u) \in \mathbb{R}^4$ whose coordinate norms fluctuate by $> 40\%$ along every orbit (Figure 5a; full equations in Appendix D). Across 100 whole orbits split into 60 train, 20 validation, and 20 held-out test orbits (5 seeds), Table 4 and Figure 5 confirm both parts of Theorem 3:

Table 4: **Five-seed 4D Coupled Nonlinear Oscillator ($K = 2$) toroidal invariant identification benchmark (20 held-out test orbits on $\mathbb{T}^2$, $T = 120$).** "1-Scalar Readout R^2" predicts (I_1, I_2) from total norm $\|\bar{\boldsymbol{\rho}}\|_2 \in \mathbb{R}$; "2-Block Readout R^2" predicts (I_1, I_2) from the 2D block-radius vector $(\bar{\rho}_1, \bar{\rho}_2) \in \mathbb{R}^2$.

Architecture ($d_z = 4 = 2 \oplus 2$)	**Block Radial** $\mathrm{CV}_{\mathcal{O}}$ (↓)	**1-Scalar Readout** R^2	**2-Block Readout** R^2 (↑)	**Rollout MSE** (↓)
2-Block Toroidal Geodesic SkipNorm ($K = 2$)	$\mathbf{0.0187 \pm 0.0029}$	0.1461 ± 0.0209	$\mathbf{0.9993 \pm 0.0002}$	$\mathbf{0.0018 \pm 0.0005}$
Single-Sphere Geodesic SkipNorm ($K = 1$)	0.3213 ± 0.0566	0.1750 ± 0.0176	0.5199 ± 0.2966	0.0031 ± 0.0011
Unconstrained Autoencoder ($d_z = 4$)	0.3529 ± 0.0285	0.1590 ± 0.0074	0.3263 ± 0.2121	0.0025 ± 0.0008

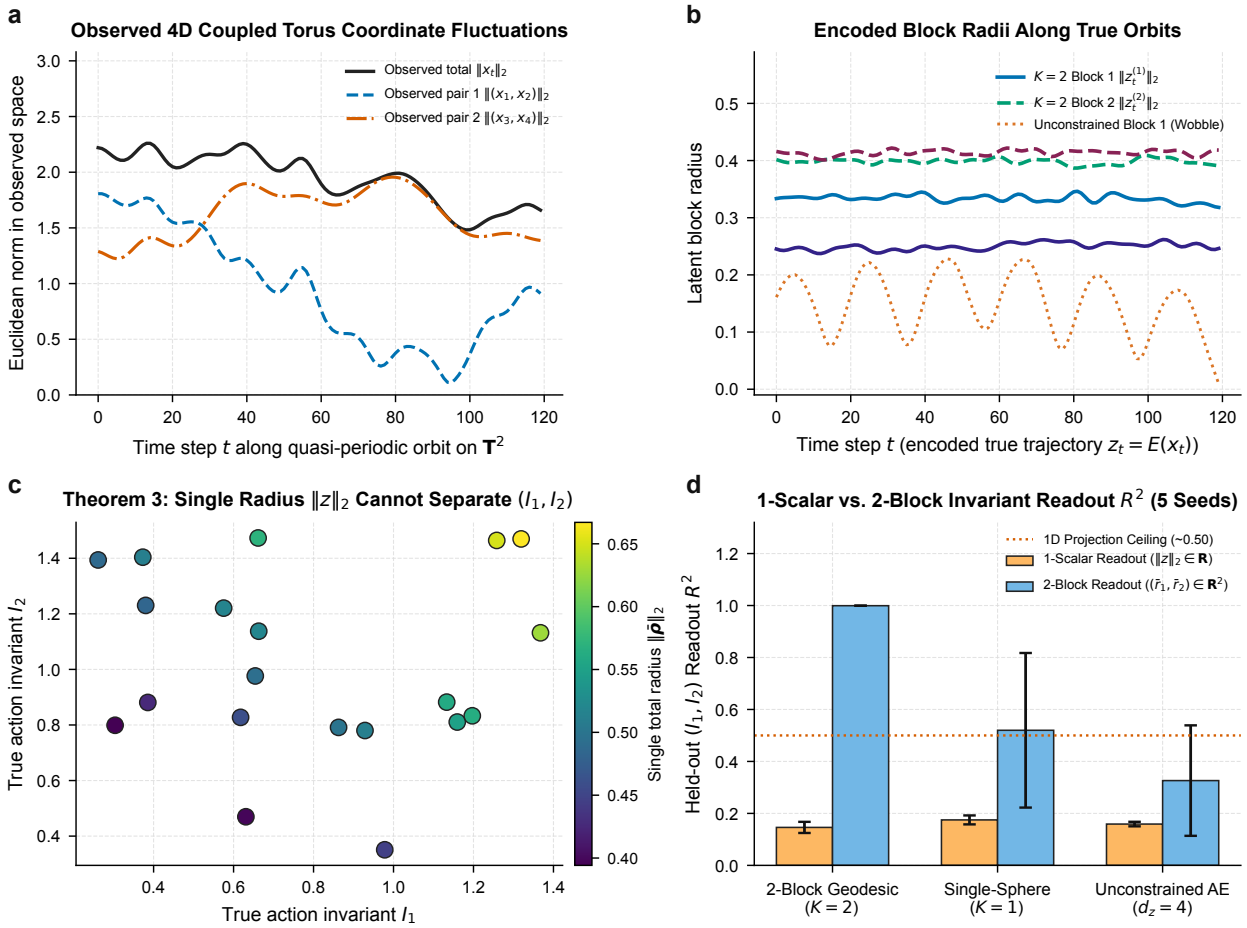


Figure 5: **Multi-invariant toroidal foliation on 4D coupled nonlinear oscillators ($K = 2$, Theorem 3). (a)** Observed coordinate norms along a single quasi-periodic test trajectory on $\mathbb{T}^2$: nonlinear cubic mixing causes both pair norms $\|(x_1, x_2)\|_2, \|(x_3, x_4)\|_2$ and the total norm $\|x_t\|_2$ to oscillate by $> 40\%$. **(b)** Encoded latent block radii $\left\|z_t^{(1)}\right\|_2, \left\|z_t^{(2)}\right\|_2$ along true test trajectories $z_t = E(x_t)$: 2-Block Toroidal Geodesic SkipNorm ($K = 2$) flattens both block radii into time-invariant constants ($\mathrm{CV}_{\mathcal{O}} = 0.0187$), whereas the Unconstrained AE exhibits severe intra-orbit wobble ($\mathrm{CV}_{\mathcal{O}} = 0.3529$). **(c)** Held-out test orbits in the true action plane (I_1, I_2) colored by single total radius $\|\bar{\boldsymbol{\rho}}\|_2$: level contours of $\|\bar{\boldsymbol{\rho}}\|_2$ are 1D curves in $\mathbb{R}^2$ that cannot separate (I_1, I_2) (Theorem 3(a)). **(d)** Held-out (I_1, I_2) readout R^2 across 5 seeds: 1-scalar readout from $\|z\|_2$ is capped below the 0.50 projection ceiling for all models, whereas 2-block readout from $(\bar{\rho}_1, \bar{\rho}_2)$ under 2-Block Toroidal Geodesic SkipNorm achieves $R^2 = 0.9993 \pm 0.0002$.

1. **Verification of the Single-Scalar Ceiling (Theorem 3(a)):** Predicting $(I_1, I_2) \in \mathbb{R}^2$ from any single scalar radius $\|z\|_2 \in \mathbb{R}$ is upper-bounded by the 1D projection ceiling (≤ 0.50), yielding $R^2 \in [0.146, 0.175]$ across all models (Figure 5c,d). Moreover, Single-Sphere Geodesic SkipNorm

($K = 1$) only constrains $\rho_1^2 + \rho_2^2$, allowing energy to slosh between the two 2D subspaces (`Block CV` $= 0.3213$, 2-Block Readout $R^2 = 0.520 \pm 0.297$).

2. **2-Block Toroidal Action Recovery (Theorem 3(b)):** 2-Block Toroidal Geodesic SkipNorm ($K = 2$) drives within-orbit block radial variation down to $\mathrm{CV}_{\mathcal{O}} = 0.0187 \pm 0.0029$ (Figure 5b) and recovers both hidden action invariants on held-out test orbits with **2-Block Readout** $R^2 = 0.9993 \pm 0.0002$ (Figure 5d).

8 Discussion, Limitations, and Conclusion

Our theoretical and empirical results establish both the utility and the exact boundaries of norm-transferring skip connections:

1. **Conditional Nature of Radial Foliation:** Theorems 1 and 3 are conditional on exact intertwining ($E \circ \Phi_{\Delta t} = g \circ E$), topological orbit density within connected level sets, and global encoder injectivity. In practice, finite-horizon training achieves approximate intertwining ($\mathrm{CV}_{\mathcal{O}}(h) \approx 0.02$–$0.04$), and systems with resonant non-dense orbits (Figure 4c) or chaotic non-integrable regions do not satisfy (H2).

2. **Multi-Basin Phase Spaces and Unknown Invariant Count K:** As proved in Theorem 2, global homeomorphisms cannot map disjoint sub-barrier energy wells onto concentric origin-centered spheres; handling multi-basin topologies requires either atlas/chart-indexed encoders or non-injective basin folding. Similarly, K-Block SkipNorm requires specifying the number of blocks K; automatically pruning redundant blocks via radial variance regularization is a promising direction for future work.

3. **Dissipative and Open Systems:** Strict norm preservation ($\|g(z)\|_2 = \|z\|_2$) is tailored to conservative or unitary flows. Extending Geodesic SkipNorm with a learned radial gating factor $\hat{y} = \gamma_\theta(x) \|x\|_2 (x + \alpha f_\perp(x)) / \|x + \alpha f_\perp(x)\|_2$ enables controlled radial contraction or expansion for dissipative dynamics while preserving the exact cotangent pullback structure up to $\gamma_\theta(x)$.

Broader Impact Statement

This work investigates fundamental mathematical and architectural properties of normalization layers and dynamical autoencoders. It carries no direct societal or ethical risks beyond those common to foundational machine learning and scientific computing methods.

Reproducibility Statement

All proofs are provided in Appendices A–C. Complete JAX (float64) source code for all layers, unit tests, dataset generators with SHA-256 split manifests, 5-seed training scripts, and figure export manifests are included in the repository.

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A Proofs of Differential Geometry Propositions and Counterexamples

A.1 Proof of Proposition 1 (Exact Jacobian Geometry of Euclidean SkipNorm)

Proof. Let $x \in \mathbb{R}^{d_{\text{in}}} \setminus \{0\}$ with $r = \|x\|_2 > 0$ and $u_{\text{in}} = x/r$. Since $f : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^{d_{\text{out}}}$ is C^1 near x and $y = f(x) \neq 0$, the map $u_{\text{out}}(x) = f(x)/\|f(x)\|_2$ is C^1 near x with derivative

$$D_x u_{\text{out}}(x) = \frac{1}{\|f(x)\|_2} \left(I_{d_{\text{out}}} - u_{\text{out}}(x) u_{\text{out}}(x)^\top\right) D_x f(x). \tag{19}$$

Notice that $u_{\text{out}}(x)^\top D_x u_{\text{out}}(x) = 0$, so $\text{im}(D_x u_{\text{out}}(x)) \subseteq u_{\text{out}}(x)^\perp$. By the product rule applied to $F(x) = r(x) u_{\text{out}}(x)$ with $D_x r(x) = u_{\text{in}}^\top$,

$$J(x) = D_x F(x) = u_{\text{out}}(x) D_x r(x) + r(x) D_x u_{\text{out}}(x) = u_{\text{out}} u_{\text{in}}^\top + r D_x u_{\text{out}}(x). \tag{20}$$

Inserting the orthogonal decomposition $I_{d_{\text{in}}} = u_{\text{in}} u_{\text{in}}^\top + P_{u_{\text{in}}^\perp}$ after $r D_x u_{\text{out}}(x)$ yields

$$J(x) = u_{\text{out}} u_{\text{in}}^\top + \underbrace{r(D_x u_{\text{out}}(x)) u_{\text{in}}}_{B(x) \in u_{\text{out}}^\perp} u_{\text{in}}^\top + \underbrace{\left(r D_x u_{\text{out}}(x)\big|_{u_{\text{in}}^\perp}\right)}_{A_\perp(x) : u_{\text{in}}^\perp \to u_{\text{out}}^\perp} P_{u_{\text{in}}^\perp}, \tag{21}$$

which establishes Eq. (4) in part (a).

For part (b), left-multiplying $J(x)^\top = u_{\text{in}} u_{\text{out}}^\top + u_{\text{in}} B(x)^\top + P_{u_{\text{in}}^\perp} A_\perp(x)^\top$ onto u_{out} and using $B(x) \in u_{\text{out}}^\perp \implies B(x)^\top u_{\text{out}} = 0$ and $\text{im}(A_\perp(x)) \subseteq u_{\text{out}}^\perp \implies A_\perp(x)^\top u_{\text{out}} = 0$ gives

$$J(x)^\top u_{\text{out}} = u_{\text{in}}(u_{\text{out}}^\top u_{\text{out}}) + 0 + 0 = u_{\text{in}}. \tag{22}$$

Hence $\left\|J(x)^\top u_{\text{out}}\right\|_2 = \|u_{\text{in}}\|_2 = 1$, and $\sigma_{\max}(J(x)) = \left\|J(x)^\top\right\|_{\text{op}} \geq \left\|J(x)^\top u_{\text{out}}\right\|_2 / \|u_{\text{out}}\|_2 = 1$. By induction over L layers with radial unit vectors $u_\ell = x_\ell/\|x_\ell\|_2$, $J_{\text{total}}^\top u_L = J_1^\top \cdots J_L^\top u_L = J_1^\top \cdots J_{L-1}^\top u_{L-1} = \cdots = u_0$.

For part (c), right-multiplying $J(x)$ onto u_{in} and using $P_{u_{\text{in}}^\perp} u_{\text{in}} = 0$ gives $J(x) u_{\text{in}} = u_{\text{out}} + B(x)$. Since $u_{\text{out}} \perp B(x)$, $\|J(x) u_{\text{in}}\|_2^2 = 1 + \|B(x)\|_2^2$, so $J(x) u_{\text{in}} = u_{\text{out}} \iff B(x) = 0$. If $f(\lambda x) = \lambda^\gamma f(x)$ for $\lambda > 0$, then $u_{\text{out}}(\lambda x) = u_{\text{out}}(x)$, so differentiating with respect to λ at $\lambda = 1$ gives $B(x) = r(D_x u_{\text{out}}(x)) u_{\text{in}} = (D_x u_{\text{out}}(x)) x = 0$. When $B(x) = 0$, $[J(x)] = \text{diag}(1, \tilde{A})$, so $J(x)^\top J(x)$ has block-diagonal form $\text{diag}(1, \tilde{A}^\top \tilde{A})$ with singular values $\{1\} \cup \sigma(\tilde{A})$.

For part (d)(i), let $f(x) = R_{\log\|x\|_2} x$ on $\mathbb{R}^2 \setminus \{0\}$. In polar coordinates (r, θ), $F(r, \theta) = (r\cos(\theta + \log r), r\sin(\theta + \log r))^\top$. At $x = e_1$ $(r = 1, \theta = 0)$, $\partial F/\partial r = (\cos 0 - \sin 0, \sin 0 + \cos 0)^\top = (1, 1)^\top$ and $\partial F/\partial \theta = (-\sin 0, \cos 0)^\top = (0, 1)^\top$, so $J(e_1) = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$. The eigenvalues of $J(e_1)^\top J(e_1) = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ satisfy $\lambda^2 - 3\lambda + 1 = 0 \implies \lambda = \frac{3 \pm \sqrt{5}}{2} = \varphi^2, \varphi^{-2}$, so $\sigma(J(e_1)) = \{\varphi, \varphi^{-1}\}$. For part (d)(ii), $F(t) = t(\cos\log t, \sin\log t)^\top$ has $F'(1) = (1, 1)^\top$, whose singular value is $\|F'(1)\|_2 = \sqrt{2}$, while $F'(1)^\top u_{\text{out}}(1) = (1, 1)(1, 0)^\top = 1 = u_{\text{in}}$. □

A.2 Proof of Proposition 2 (Exact Rank and Stabilized Normalizer Spectra)

Proof. **Part (a):** By Proposition 1(a), $[J_{\text{SkipNorm}}(x)] = \left(\begin{smallmatrix} 1 & 0 \\ b & \tilde{A} \end{smallmatrix}\right)$. Subtracting b times the first row from the remaining rows via the invertible row operation $\left(\begin{smallmatrix} 1 & 0 \\ -b & I_{d_{\text{out}}-1} \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 0 \\ b & \tilde{A} \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & \tilde{A} \end{smallmatrix}\right)$ proves $\text{rank}(J_{\text{SkipNorm}}(x)) = 1 + \text{rank}(\tilde{A}) = 1 + \text{rank}(A|_T)$. For unstabilized RMSNorm ($\varepsilon = 0$), $D_x(\sqrt{d} f(x)/\|f(x)\|_2) = \frac{\sqrt{d}}{\|y\|_2} P_\perp A$, whose rank is $\text{rank}(P_\perp A)$. For the 2×2 counterexample at $x = e_1$ ($u_{\text{in}} = e_1$) with $f(x) = (1 + x_2, 0)^\top$, $f(e_1) = e_1$ ($u_{\text{out}} = e_1$) and $A = D_x f(e_1) = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has $\text{rank}(A) = 1$, yet $P_\perp = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right) \implies P_\perp A = 0 \implies A|_T = 0$, so $\text{rank}(J_{\text{SkipNorm}}(e_1)) = 1 + 0 = 1$.

Part (b): Differentiating $g_\varepsilon(y) = \sqrt{d}(r^2 + d\varepsilon)^{-1/2} y$ gives

$$D_y g_\varepsilon(y) = \frac{\sqrt{d}}{\sqrt{r^2+d\varepsilon}} I_d - \frac{\sqrt{d}}{(r^2+d\varepsilon)^{3/2}} yy^\top = \frac{\sqrt{d}}{\sqrt{r^2+d\varepsilon}}(I_d - uu^\top) + \left(\frac{\sqrt{d}}{\sqrt{r^2+d\varepsilon}} - \frac{\sqrt{d}r^2}{(r^2+d\varepsilon)^{3/2}}\right) uu^\top$$
$$= \frac{\sqrt{d}}{\sqrt{r^2+d\varepsilon}}(I_d - uu^\top) + \frac{d^{3/2}\varepsilon}{(r^2+d\varepsilon)^{3/2}} uu^\top. \tag{23}$$

Since $(I_d - uu^\top)$ and $uu^\top$ are orthogonal symmetric projectors onto $u^\perp$ ($\dim = d-1$) and $\mathrm{span}(u)$ ($\dim = 1$), the eigenvalues (and singular values) are immediately given by Eq. (8).

Part (c): Let $P_{\mathbf{1}^\perp} = I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top$. Since $\mathrm{LayerNorm}_\varepsilon(y) = \mathrm{RMSNorm}_\varepsilon(P_{\mathbf{1}^\perp} y)$ and $P_{\mathbf{1}^\perp} u_c = u_c$, the chain rule gives

$$D_y \,\mathrm{LayerNorm}_\varepsilon(y) = \left[\frac{\sqrt{d}}{\sqrt{r_c^2+d\varepsilon}}(I_d - u_c u_c^\top) + \frac{d^{3/2}\varepsilon}{(r_c^2+d\varepsilon)^{3/2}} u_c u_c^\top\right] P_{\mathbf{1}^\perp}$$
$$= \frac{\sqrt{d}}{\sqrt{r_c^2+d\varepsilon}}(P_{\mathbf{1}^\perp} - u_c u_c^\top) + \frac{d^{3/2}\varepsilon}{(r_c^2+d\varepsilon)^{3/2}} u_c u_c^\top, \tag{24}$$

yielding singular values $\sigma_{\tan}(r_c;\varepsilon)$ on $\mathrm{span}(\mathbf{1}, u_c)^\perp$ ($\dim = d-2$), $\sigma_{\mathrm{rad}}(r_c;\varepsilon)$ on $\mathrm{span}(u_c)$ ($\dim = 1$), and 0 on $\mathrm{span}(\mathbf{1})$ ($\dim = 1$). □

A.3 Proof of Proposition 3 (Tangent-Projected Geodesic SkipNorm)

Proof. By definition, $f_\perp(x) = (I_d - u_{\mathrm{in}} u_{\mathrm{in}}^\top) f(x)$ satisfies $\langle f_\perp(x), x\rangle = 0$. Therefore $\|x + \alpha f_\perp(x)\|_2^2 = \|x\|_2^2 + \alpha^2 \|f_\perp(x)\|_2^2 \geq \|x\|_2^2 > 0$ for all $\alpha \in \mathbb{R}$. Taking the inner product of $\hat{y} = \|x\|_2 (x + \alpha f_\perp(x))/\sqrt{\|x\|_2^2 + \alpha^2 \|f_\perp(x)\|_2^2}$ with $x/\|x\|_2^2$ immediately yields Eq. (12). Expanding $(1 + \alpha^2 \|f_\perp(x)\|_2^2 / \|x\|_2^2)^{-1/2} = 1 - \frac{\alpha^2 \|f_\perp(x)\|_2^2}{2\|x\|_2^2} + \mathcal{O}(\alpha^4)$ yields the Taylor expansion in part (b). □

B Proofs of Conditional Radial Foliation and Resonant Oscillator Control

B.1 Proof of Theorem 1 (Conditional Radial Foliation)

Proof. **Part (a):** Fix $c \in (c_{\min}, c_{\max})$ and let $x_0 \in \mathcal{O}_c$ be a point whose forward orbit $\Gamma(x_0) = \{\Phi_{\Delta t}^k(x_0) : k \in \mathbb{N}\}$ is dense in $\mathcal{O}_c$ by (H2). By (H1), $E(\Phi_{\Delta t}^k(x_0)) = g^k(E(x_0))$ and $\|g(z)\|_2 = \|z\|_2$, so induction on k gives $\left\|E(\Phi_{\Delta t}^k(x_0))\right\|_2 = \|E(x_0)\|_2 =: \psi(c)$ for all $k \in \mathbb{N}$. Since $E : U \to \mathbb{R}^{d_z}$ and $\|\cdot\|_2 : \mathbb{R}^{d_z} \to \mathbb{R}_{\geq 0}$ are continuous, the composite function $h(x) = \|E(x)\|_2$ is continuous on $\mathcal{O}_c$. Because h is constant on the dense subset $\Gamma(x_0) \subseteq \mathcal{O}_c$, $h(x) = \psi(c)$ for all $x \in \overline{\Gamma(x_0)} = \mathcal{O}_c$, proving Eq. (14).

Now suppose every $c \in (c_{\min}, c_{\max})$ is a regular value of $I \in C^1(U, \mathbb{R})$. Fix $c_0 \in (c_{\min}, c_{\max})$ and pick $x_0 \in \mathcal{O}_{c_0}$. Since $\nabla I(x_0) \neq 0$, the smooth curve $\gamma(s) = x_0 + s\nabla I(x_0)/\|\nabla I(x_0)\|_2^2$ satisfies $(I \circ \gamma)'(0) = 1 \neq 0$. By the Inverse Function Theorem on $\mathbb{R}$, $I \circ \gamma$ is a C^1 diffeomorphism from an open neighborhood $(-\delta, \delta)$ onto an open neighborhood $(c_0 - \epsilon, c_0 + \epsilon) \subseteq (c_{\min}, c_{\max})$ with continuous inverse $\tau : (c_0 - \epsilon, c_0 + \epsilon) \to (-\delta, \delta)$. Then $\sigma(c) \coloneqq \gamma(\tau(c))$ is a continuous local section satisfying $I(\sigma(c)) = c$, so $\sigma(c) \in \mathcal{O}_c$ and $\psi(c) = \|E(\sigma(c))\|_2$ is a composition of continuous maps, hence continuous at c_0.

Part (b): Assume (H3) ($d_z = N \geq 2$, $\mathcal{O}_c$ regular, compact, and connected, and $E : U \to \mathbb{R}^N$ continuous and injective). First, if $\psi(c) = 0$ for some c, then $E(\mathcal{O}_c) = \{0\}$, contradicting injectivity of E since $\mathcal{O}_c$ is a compact $(N-1)$-manifold with $N \geq 2$ (hence infinite). Thus $\psi(c) > 0$. By the Regular Value Theorem, $\mathcal{O}_c$ is a compact $(N-1)$-dimensional topological manifold without boundary, and $\mathbb{S}^{N-1}(\psi(c))$ is a connected $(N-1)$-dimensional manifold without boundary ($N \geq 2$). The restriction $E|_{\mathcal{O}_c} : \mathcal{O}_c \to \mathbb{S}^{N-1}(\psi(c))$ is continuous and injective. By Brouwer's Invariance of Domain for manifolds of equal dimension $(N-1)$ (Brouwer, 1911; Hatcher, 2002), $E(\mathcal{O}_c)$ is open in $\mathbb{S}^{N-1}(\psi(c))$. Since $\mathcal{O}_c$ is compact and $\mathbb{S}^{N-1}(\psi(c))$ is Hausdorff, $E(\mathcal{O}_c)$ is also compact and hence closed in $\mathbb{S}^{N-1}(\psi(c))$. Because $\mathbb{S}^{N-1}(\psi(c))$ is connected for

$N \geq 2$ and $E(\mathcal{O}_c) \neq \emptyset$, the clopen subset $E(\mathcal{O}_c)$ must equal the entire sphere $\mathbb{S}^{N-1}(\psi(c))$. A continuous bijection from a compact space $\mathcal{O}_c$ onto a Hausdorff space $\mathbb{S}^{N-1}(\psi(c))$ is a homeomorphism.

Finally, for any $c_1 \neq c_2$ in $(c_{\min}, c_{\max})$, $\mathcal{O}_{c_1} \cap \mathcal{O}_{c_2} = \emptyset$. By global injectivity of E on U, $E(\mathcal{O}_{c_1}) \cap E(\mathcal{O}_{c_2}) = \mathbb{S}^{N-1}(\psi(c_1)) \cap \mathbb{S}^{N-1}(\psi(c_2)) = \emptyset$, which implies $\psi(c_1) \neq \psi(c_2)$. Thus $\psi : (c_{\min}, c_{\max}) \to \mathbb{R}_{>0}$ is continuous and injective on an open interval, hence strictly monotone, mapping $(c_{\min}, c_{\max})$ homeomorphically onto an open interval $(\psi_{\min}, \psi_{\max}) \subseteq \mathbb{R}_{>0}$. □

B.2 Proof of the Resonant $1:1$ Harmonic Oscillator Control

Proposition 4 (Non-Foliation Under Non-Dense Orbits)**.** *On $\mathbb{R}^4$ with coordinates $x = (q_1, p_1, q_2, p_2)^\top$, let $\Phi_{\Delta t}(x) = \mathrm{diag}(R_{\Delta t}, R_{\Delta t})x$ be the flow of the isotropic $1:1$ harmonic oscillator with Hamiltonian $H(x) = \frac{1}{2}(q_1^2+p_1^2+q_2^2+p_2^2) = H_1(x)+H_2(x)$. Let $E(x) = (\sqrt{2}q_1, \sqrt{2}p_1, q_2, p_2)^\top$ and $g(z) = \mathrm{diag}(R_{\Delta t}, R_{\Delta t})z \in \mathrm{SO}(4)$. Then E is a global linear diffeomorphism, $E \circ \Phi_{\Delta t} = g \circ E$ holds identically on $\mathbb{R}^4$, and $\left\|E(\Phi^k_{\Delta t}(x))\right\|_2 = \|E(x)\|_2$ for all $k \in \mathbb{Z}$, yet on every energy shell $\mathcal{O}_c = \{x : H(x) = c\} \cong \mathbb{S}^3(\sqrt{2c})$ $(c > 0)$, $\|E(\mathcal{O}_c)\|_2 = [\sqrt{2c}, 2\sqrt{c}]$ is a non-degenerate interval.*

Proof. Since $\mathrm{diag}(\sqrt{2}I_2, I_2)$ commutes with $\mathrm{diag}(R_{\Delta t}, R_{\Delta t})$, $E(\Phi_{\Delta t}(x)) = g(E(x))$ holds identically. Along any trajectory, $H_1(x_t) = \frac{1}{2}(q_1(t)^2 + p_1(t)^2)$ and $H_2(x_t) = \frac{1}{2}(q_2(t)^2 + p_2(t)^2)$ are individually constant, so $\|E(x_t)\|_2 = \sqrt{4H_1(x_0) + 2H_2(x_0)} = \sqrt{2c(1+\eta)}$ where $\eta = H_1(x_0)/c \in [0,1]$ ranges over $[0,1]$ on $\mathcal{O}_c$. □

C Proofs of Topological Obstructions and Toroidal Generalization

C.1 Proof of Theorem 2 (Jordan–Brouwer Multi-Well Obstruction) and Translated Control

Proof. **Part (a) and (b):** Since $\mathcal{O}^{(k)}_{c_k} \cong \mathbb{S}^{N-1}$ $(k = 1, 2)$ and $E : U \to \mathbb{R}^N$ is continuous and injective, each image $\Sigma_k := E(\mathcal{O}^{(k)}_{c_k})$ is an embedded $(N-1)$-sphere in $\mathbb{R}^N$. By the Jordan–Brouwer Separation Theorem (Hatcher, 2002), $\mathbb{R}^N \setminus \Sigma_k$ has exactly two connected components: a bounded interior $\mathrm{Int}(\Sigma_k)$ and an unbounded exterior $\mathrm{Ext}(\Sigma_k)$. By Brouwer's Invariance of Domain, E is an open map from $U \subseteq \mathbb{R}^N$ to $\mathbb{R}^N$. Since $B_k := \mathrm{Int}(\mathcal{O}^{(k)}_{c_k}) \subset U$ is connected and bounded with $\partial B_k = \mathcal{O}^{(k)}_{c_k}$, its image $E(B_k)$ is a connected open bounded set in $\mathbb{R}^N \setminus \Sigma_k$ with $\partial E(B_k) = \Sigma_k$, which implies $E(B_k) = \mathrm{Int}(\Sigma_k)$. Because $B_1 \cap B_2 = \emptyset$ and E is injective on U, $\mathrm{Int}(\Sigma_1) \cap \mathrm{Int}(\Sigma_2) = E(B_1) \cap E(B_2) = \emptyset$. Therefore, $0 \in \mathbb{R}^N$ cannot belong to both $\mathrm{Int}(\Sigma_1)$ and $\mathrm{Int}(\Sigma_2)$. Because every origin-centered sphere $\mathbb{S}^{N-1}(r)$ $(r > 0)$ has $\mathrm{Int}(\mathbb{S}^{N-1}(r)) = B_N(0, r) \ni 0$, Σ_1 and Σ_2 cannot both be origin-centered spheres.

Translated Identity Encoder Control (Part (c)(iii)): Let $K \subset \mathbb{R}^N$ be a compact domain with $R = \sup_{x \in K} \|x\|_2$ and step displacement $D = \sup_{x \in K} \|\Phi_{\Delta t}(x) - x\|_2$. For $a > R$, define $E_a(x) = ae_1 + x$ and $g(z) = z$. Then for all $x \in K$, $\|E_a(x)\|_2 \in [a - R, a + R]$, so along any orbit in K the mean radius satisfies $\bar{r} \geq a - R$ and the standard deviation satisfies $s \leq R$, giving $\mathrm{CV}_{\mathcal{O}}(h) \leq \frac{R}{a-R} = \mathcal{O}(a^{-1})$ and relative intertwining error $\frac{\|E_a(\Phi_{\Delta t}(x)) - g(E_a(x))\|_2}{\|E_a(x)\|_2} = \frac{\|\Phi_{\Delta t}(x) - x\|_2}{\|ae_1 + x\|_2} \leq \frac{D}{a-R} = \mathcal{O}(a^{-1})$. □

C.2 Proof of Theorem 3 (K-Block Toroidal Generalization)

Proof. **Part (a):** By Theorem 1(a), if $\|g(z)\|_2 = \|z\|_2$ and each $\mathcal{O}_c \cong \mathbb{T}^K$ carries a dense orbit, then $\|E(x)\|_2 = \psi(I(x))$ for a continuous scalar map $\psi : \Omega \to \mathbb{R}_{\geq 0}$ on $\Omega \subseteq \mathbb{R}^K$. Since $K \geq 2 > 1$, ψ cannot be injective on the open set $\Omega \subseteq \mathbb{R}^K$ (by Invariance of Domain, no continuous injective map exists from an open subset of $\mathbb{R}^K$ into $\mathbb{R}^1$ for $K \geq 2$); at any regular value r of $\psi \in C^1(\Omega, \mathbb{R})$, $\psi^{-1}(r)$ is a $(K-1)$-dimensional submanifold of Ω. Furthermore, for $N \geq 3$, $\pi_1(\mathbb{S}^{N-1}) = 0 \neq \mathbb{Z}^K \cong \pi_1(\mathbb{T}^K)$, and for $N = 2$, $\dim(\mathbb{S}^1) = 1 < K = \dim(\mathbb{T}^K)$, so $\mathbb{T}^K$ is never homeomorphic to $\mathbb{S}^{N-1}$.

Part (b): Under g_{block} on $\bigoplus_{k=1}^K \mathbb{R}^2$, each block norm $\rho_k(z) = \left\|z^{(k)}\right\|_2$ satisfies $\rho_k(g_{\mathrm{block}}(z)) = \rho_k(z)$. Applying Theorem 1(a) to $h_k(x) = \left\|E^{(k)}(x)\right\|_2$ yields $h_k(x) = \psi_k(I(x))$ for continuous $\psi_k : \Omega \to \mathbb{R}_{\geq 0}$, so $E(\mathcal{O}_c) \subseteq \prod_{k=1}^K \mathbb{S}^1(\psi_k(c)) \cong \mathbb{T}^K$. When E is an injective immersion and $\psi_k(c) > 0$, $E|_{\mathcal{O}_c} : \mathcal{O}_c \to$

$\prod_{k=1}^{K} \mathbb{S}^1(\psi_k(c))$ is a continuous injective map between compact connected K-manifolds without boundary; by Invariance of Domain and compactness of $\mathcal{O}_c$, $E(\mathcal{O}_c) = \prod_{k=1}^{K} \mathbb{S}^1(\psi_k(c))$. Global injectivity of E on U forces $\prod_{k=1}^{K} \mathbb{S}^1(\psi_k(c_1)) \cap \prod_{k=1}^{K} \mathbb{S}^1(\psi_k(c_2)) = \emptyset$ whenever $c_1 \neq c_2$, which implies $\Psi(c_1) \neq \Psi(c_2)$, so $\Psi : \Omega \to \mathbb{R}^K_{>0}$ is continuous and injective, hence a homeomorphism onto its open image $\Psi(\Omega) \subseteq \mathbb{R}^K_{>0}$ by Invariance of Domain on $\mathbb{R}^K$. □

D Experimental Protocols, Dataset Splits, and Hyperparameters

Whole-Orbit Disjoint Split Protocol and SHA-256 Provenance. Every benchmark dataset is partitioned strictly at the level of *whole trajectories* into disjoint training (60%), validation (20%), and test (20%) orbit sets before any windowing is performed. The exact SHA-256 hashes of all split archives in `data/splits/` (recorded in `data/splits/splits_manifest.json`) are:

- `lotka_volterra.npz` (48 train / 16 val / 16 test orbits, $T = 180$): `aeea833ff1fe...`
- `double_well_balanced.npz` (36 train / 12 val / 12 test orbits, $T = 100$): `3f717bb33045...`
- `double_well_imbalanced.npz` (36 train / 12 val / 12 test orbits, $T = 100$): `09fd2e58072d...`
- `double_well_tilted.npz` (36 train / 12 val / 12 test orbits, $T = 100$): `8d349aad3f72...`
- `coupled_torus_4d.npz` (60 train / 20 val / 20 test orbits, $T = 120$): `aa076d92d9e8...`
- `dnls_1d.npz` (36 train / 12 val / 12 test orbits, $T = 250$): `17394ed7b789...`

4D Coupled Nonlinear Oscillator Equations ($K = 2$). Latent action-angle variables $(I_1, I_2) \in [0.3, 1.5]^2$ evolve with coupled frequencies $\omega_1(I_1, I_2) = 1.0 + 0.35I_1 + 0.15I_2$ and $\omega_2(I_1, I_2) = \sqrt{2} + 0.20I_1 + 0.40I_2$, forming $u_t = (\sqrt{2I_1}\cos\theta_{1,t}, \sqrt{2I_1}\sin\theta_{1,t}, \sqrt{2I_2}\cos\theta_{2,t}, \sqrt{2I_2}\sin\theta_{2,t})^\top \in \mathbb{R}^4$. Observed states $x_t = \Phi_{\text{mix}}(u_t) \in \mathbb{R}^4$ are generated by an orthogonal cross-mode rotation $R_{\text{mix}} \in \text{SO}(4)$ followed by a cubic shear diffeomorphism $s_1 = y_1 + 0.25y_2^2 + 0.15\sin(y_3)$, $s_2 = y_2 + 0.20\cos(y_4)$, $s_3 = y_3 + 0.25y_4^2 + 0.15\sin(s_1)$, $s_4 = y_4 + 0.20\cos(s_2)$ and anisotropic scaling $\text{diag}(1.25, 0.80, 1.15, 0.85)$.

Training Hyperparameters. All models are implemented in JAX with 64-bit precision (`jax_enable_x64=True`) and trained with AdamW (lr $= 2 \times 10^{-3}$, cosine decay, weight decay 10^{-5}) across 5 random seeds $\{0, 1, 2, 3, 4\}$. Autoencoders use 2-layer residual MLP or RealNVP coupling encoders/decoders (hidden width 64) trained strictly on reconstruction loss plus multi-step latent consistency and rollout MSE over horizon $K_{\text{roll}} = 6$:

$$\mathcal{L}(\theta) = \frac{1}{K_{\text{roll}}} \sum_{k=0}^{K_{\text{roll}}} \left\| D(g^k(E(x_t))) - x_{t+k} \right\|_2^2 + \lambda_{\text{lat}} \frac{1}{K_{\text{roll}}} \sum_{k=1}^{K_{\text{roll}}} \left\| g^k(E(x_t)) - E(x_{t+k}) \right\|_2^2, \qquad \lambda_{\text{lat}} = 0.5, \quad (25)$$

with zero access to ground-truth conserved quantities $I(x)$ during training or validation.